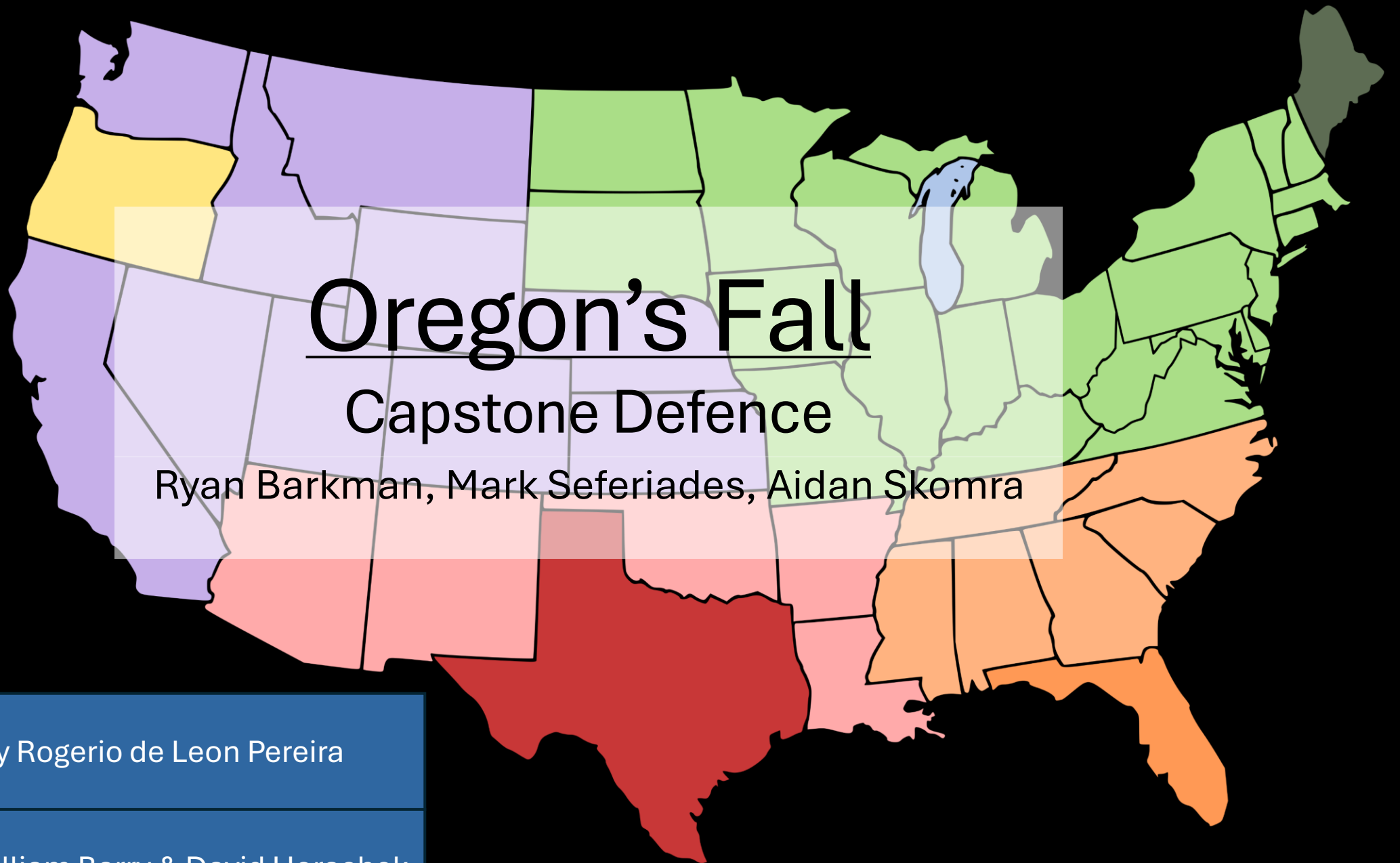




Oregon's Fall

Capstone Defence

Ryan Barkman, Mark Seferiades, Aidan Skomra

Managed By Rogerio de Leon Pereira

Monitored by William Barry & David Horachek

- <https://1.bp.blogspot.com/-HvM--V4i1RY/TjDichOkKzl/AAAAAAAAA1g/OatF69UwFwA/s400/The+Last+Stand+Union+City+walkthrough.jpg>
- <https://i0.wp.com/waytoomany.games/wp-content/uploads/2022/11/2022111614173200-010558F429C25C6688B87B1AB272C2C5.jpg?ssl=1>

Overview

- Game
 - Oregon's Fall
 - 2D Point and Click zombie shooter
 - Combat + travel
- In Game mod builder
 - Saves on cloud



Roles

Ryan

Travel Aspects
Mod Tool creation
UI Navigation
Save Systems

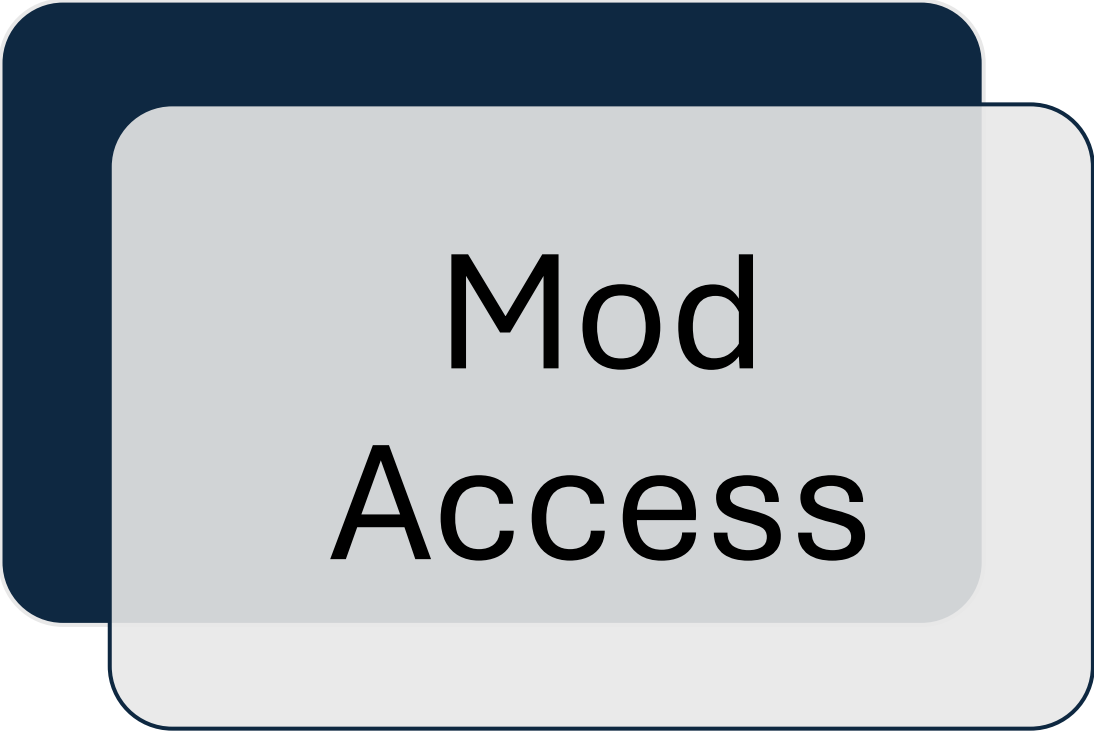
Aidan

Player Systems
Cloud Manager
Inventory

Mark

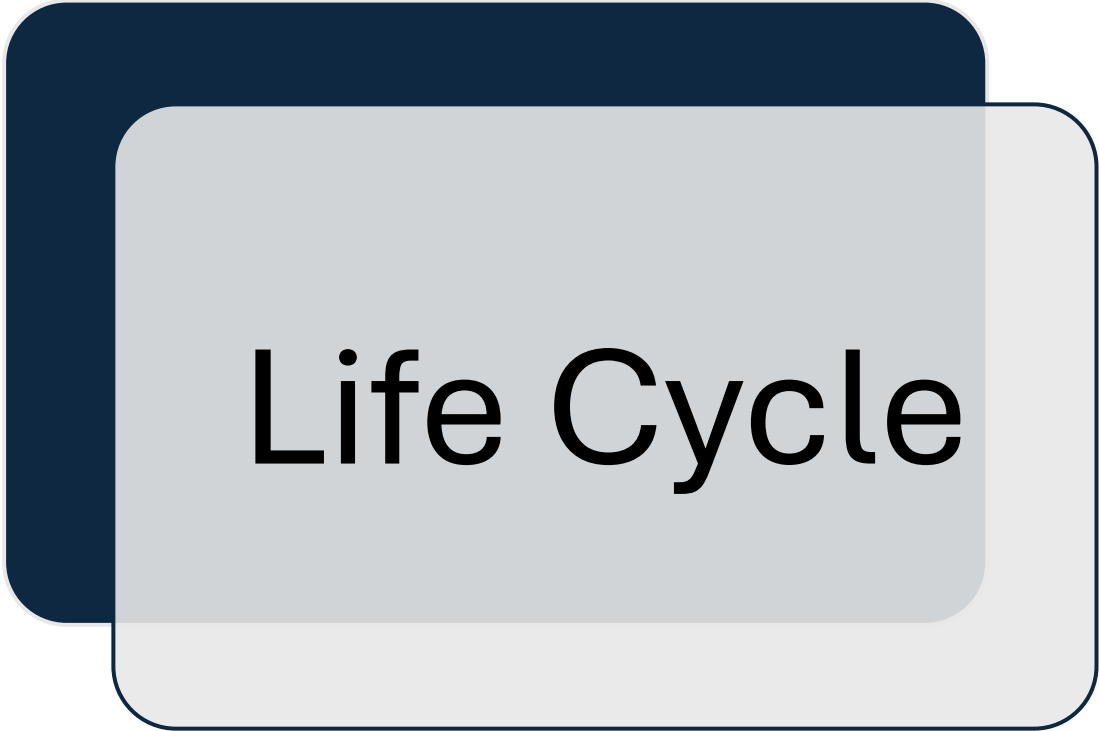
Combat
Enemy AI

Problem



Mod
Access

A diagram of a folder with a dark blue tab on the left. The folder is light gray with rounded corners. The text 'Mod Access' is centered on the front of the folder.



Life Cycle

A diagram of a folder with a dark blue tab on the left. The folder is light gray with rounded corners. The text 'Life Cycle' is centered on the front of the folder.

Why

- Mods Improve Life Cycle
- More Fun
- Community Engagement
- More players

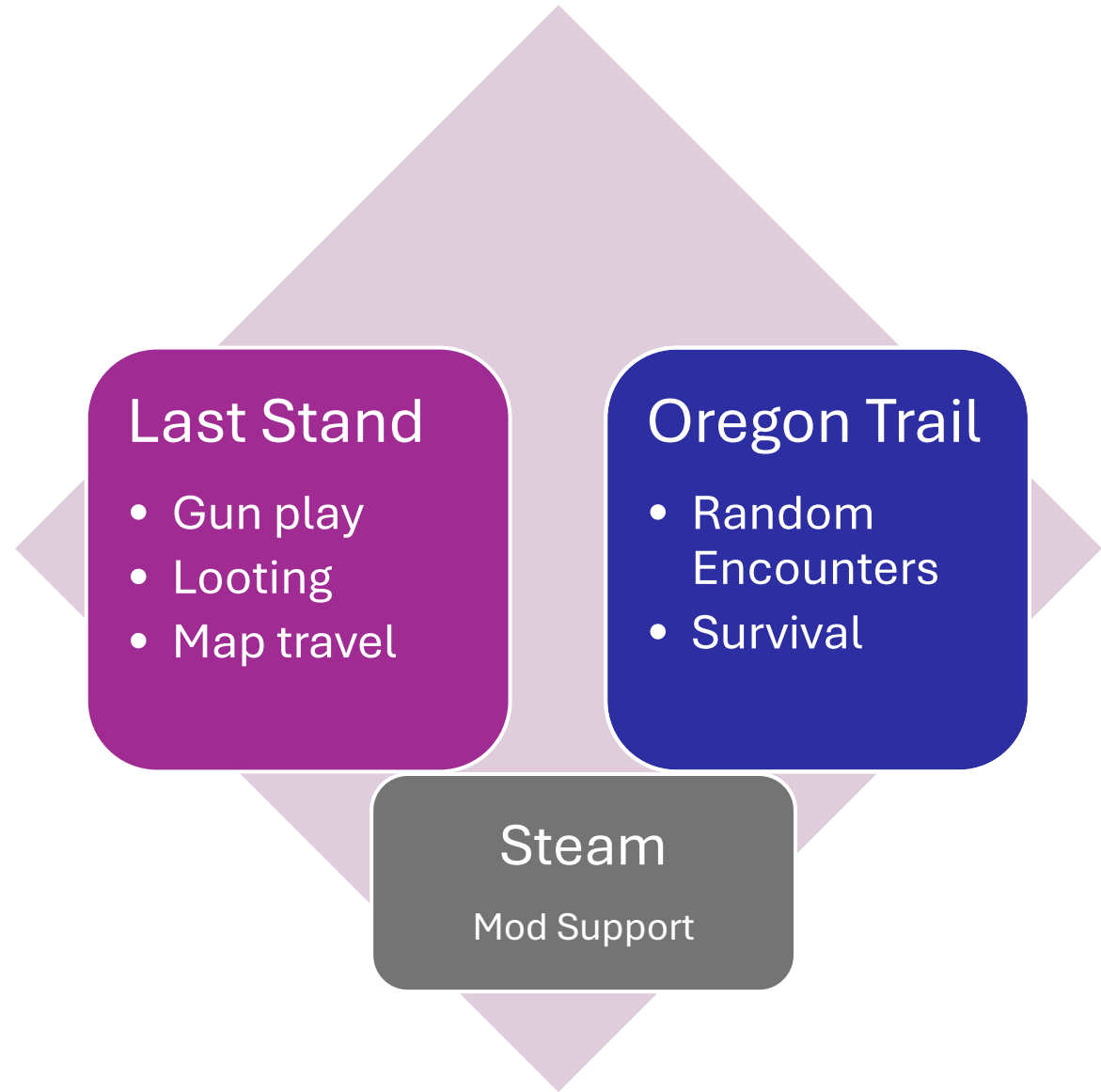
- Long Life Cycle gives good reputation
- Stays in public's eye longer

- Garry's Mod (2006) – 75th - 24,767
- Black Ops 3 (2015) – 151st (3rd in cod)
- Left 4 Dead 2 (2019) – 39th - 28,678

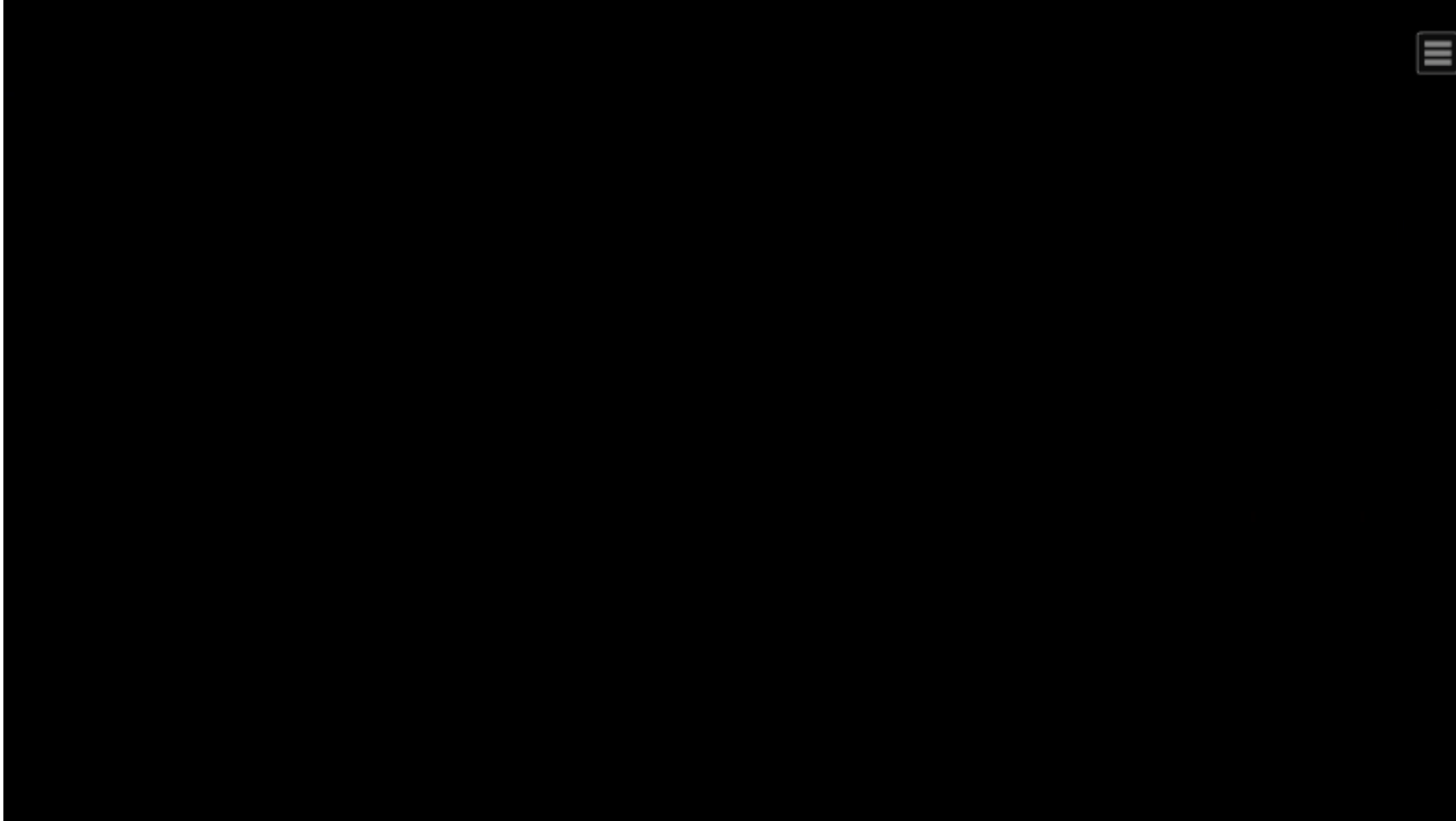
Steam DB:

<https://steamdb.info/charts/>

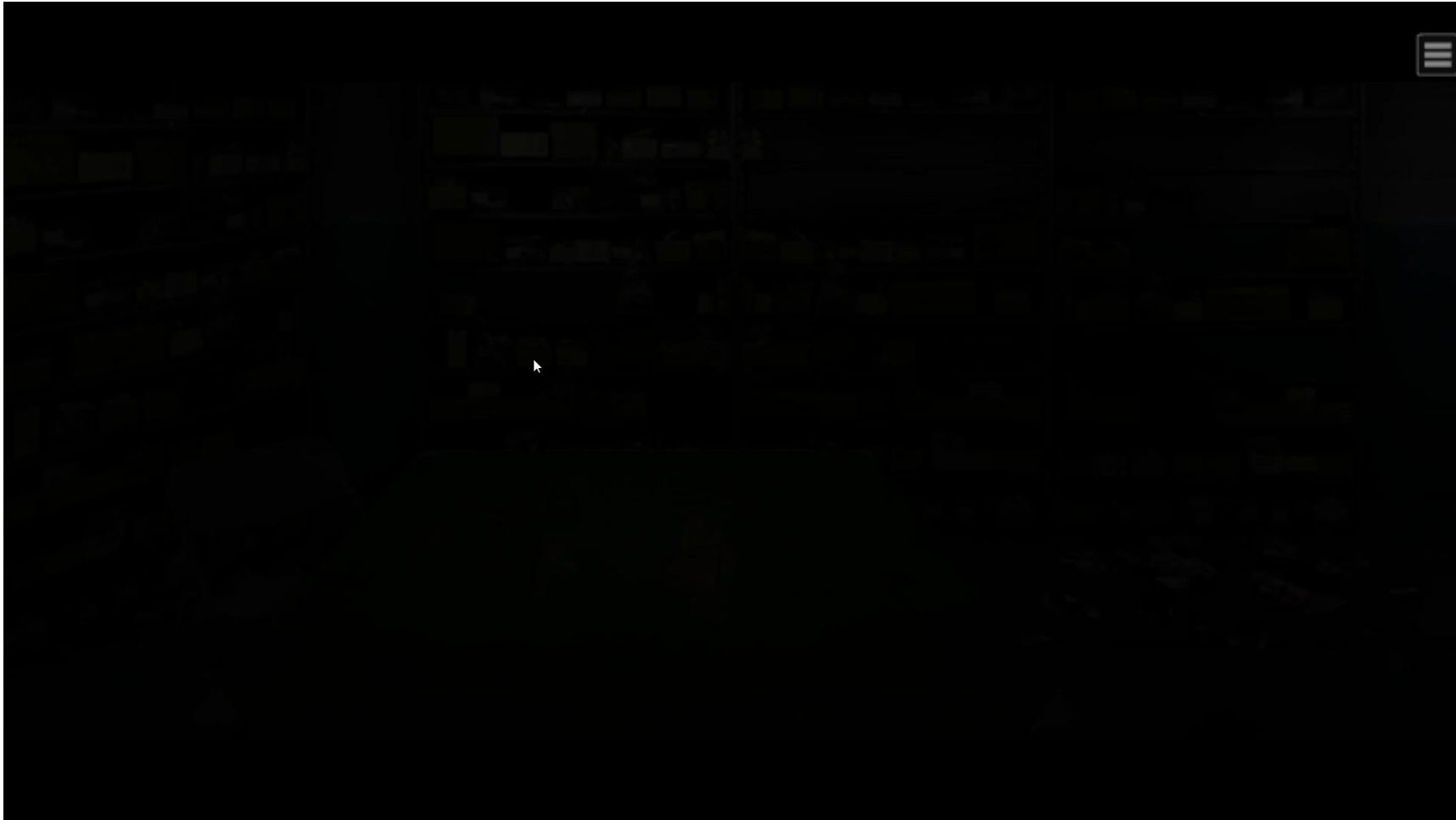
Goals & Inspirations:



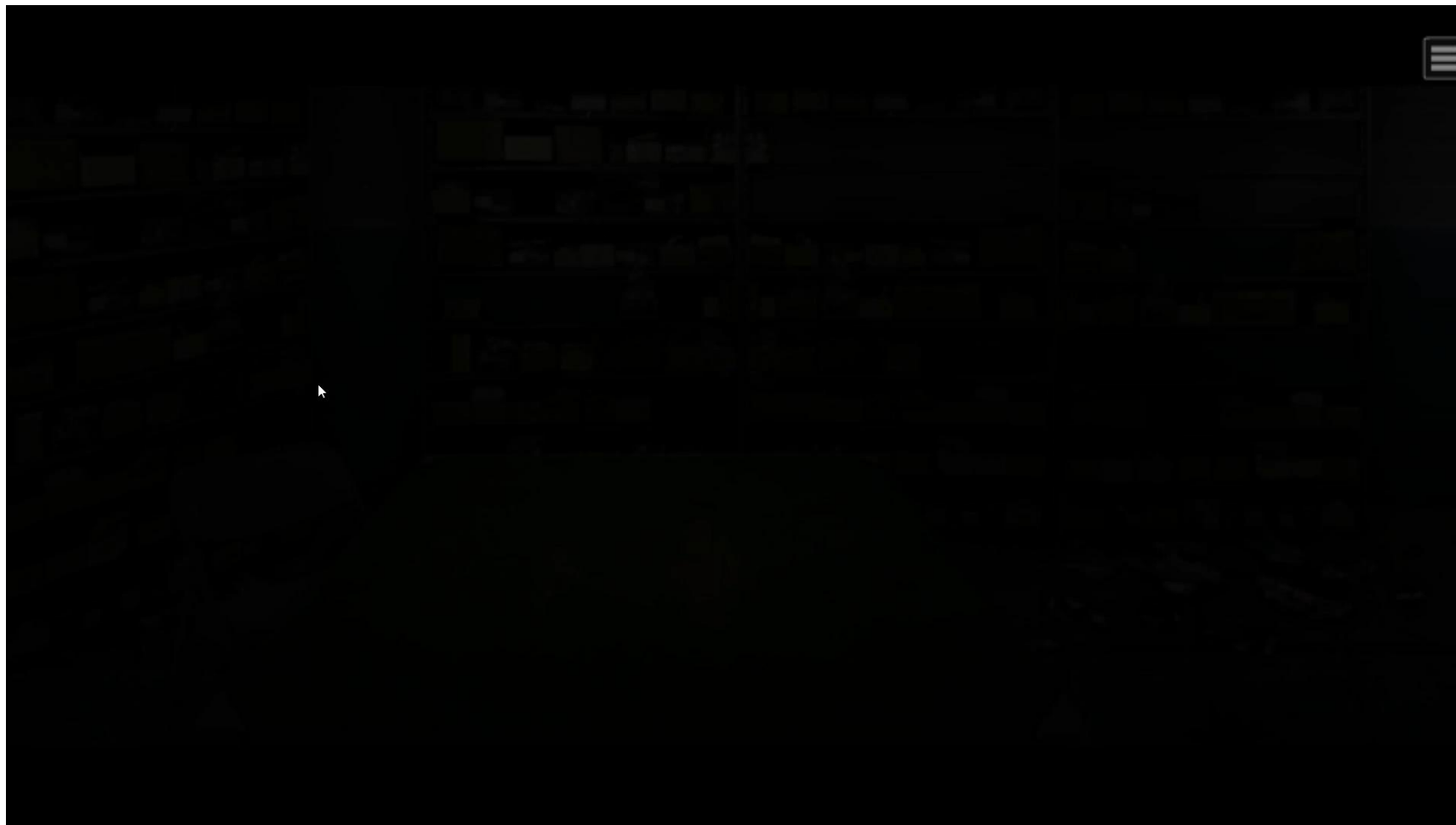
Gun Play

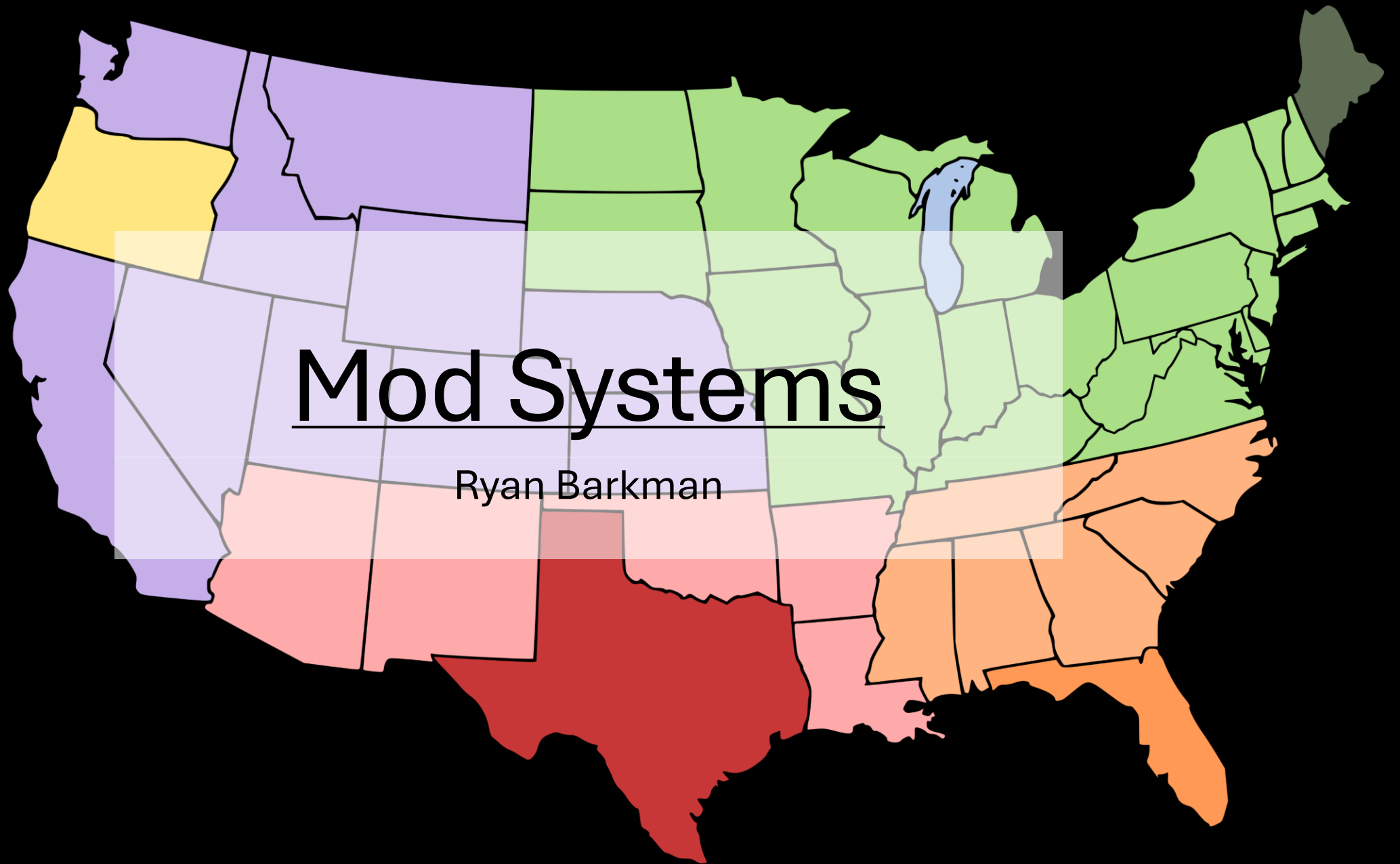


Looting



Traveling

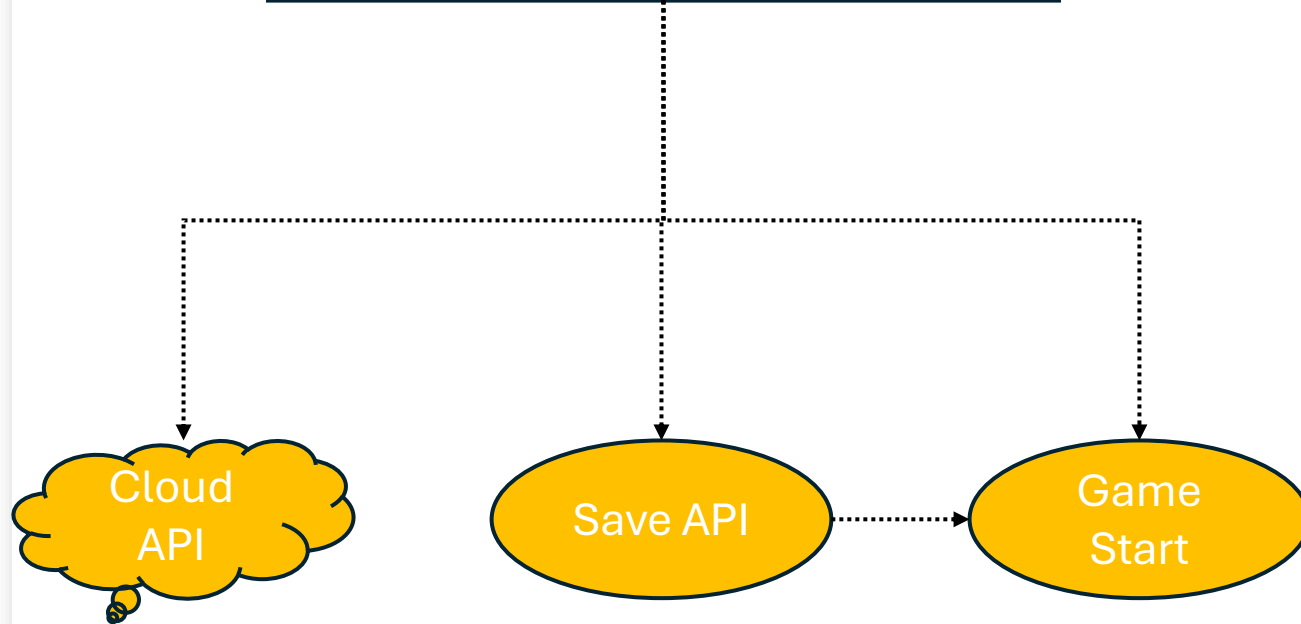




Mod Systems

Ryan Barkman

Mod Item Data	
<u>Mod Desc</u>	<u>Mod Contents</u>
Name	Mode Type
Keyword	Mod Data
Version Number	
Creator	
Description	
Photo	



Mod Items

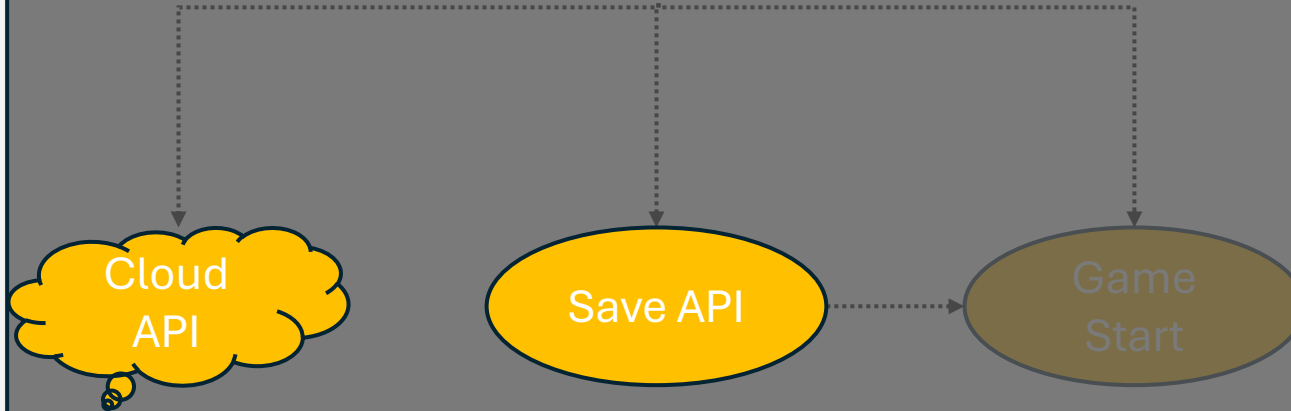
Mod Item Data

Mod Desc

Name
Keyword
Version Number
Creator
Description
Photo

Mod Contents

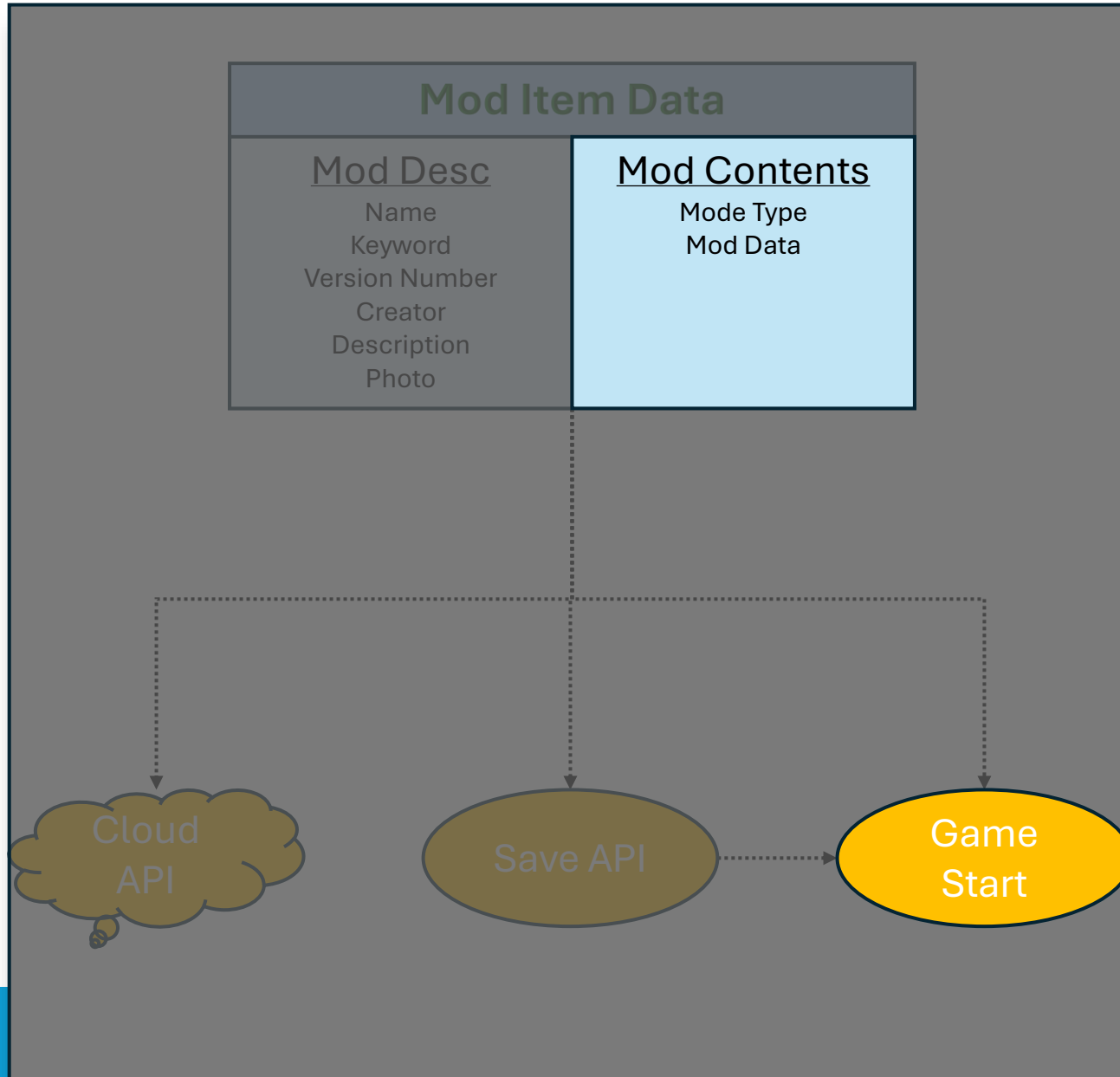
Mode Type
Mod Data



Mod Desc

Handles Classification

- Cloud And Save API use name as a search parameter
- Keyword allows Update and Delete on the cloud
- Version Number allows checking if an item should be replaced



Mod Contents

Handles Mod Data

- Simply an array of bytes

Handles type of data

- Needs to properly read it
- Also helps file format

```
public static string getFileType(ModType modType)
{
    switch (modType)
    {
        case ModType.GunMod:
            return "gpak";
        case ModType.MeeleMod:
            return "mpak";
        case ModType.AmmoMod:
            return "ampak";
        case ModType.AttachMod:
            return "atpak";
        case ModType.MapMod:
            return "mapak";
    }
    return "";
}
```

Ex: Save Custom Map Data

```
using var ms = new MemoryStream();
using var bw = new BinaryWriter(ms);

byte[] bytes = Imgs.Get();
bw.Write(bytes.Length);
bw.Write(bytes);

List<GameObject> list = Walks.Get();
bw.Write(list.Count);
foreach(GameObject go in list)
{
    UiHelper.WriteTransform(bw, go.transform);
}

list = Exits.Get();
bw.Write(list.Count);
foreach (GameObject go in list)
{
    UiHelper.WriteTransform(bw, go.transform);
}

list = Builds.Get();
bw.Write(list.Count);
foreach (GameObject go in list)
{
    UiHelper.WriteTransform(bw, go.transform);
    UiHelper.WriteTransform(bw, go.GetComponent<IsBuildable>().Viewable.transform);
}

string path = StandaloneFileBrowser.SaveFilePanel("Save Your Map", "", "NewMap", ModItemContents.getFileType(ModType.MapMod));
if (!string.IsNullOrEmpty(path))
{
    File.WriteAllBytes(path, ms.ToArray());
}
```

Img

Walks

Exits

Buildings

Saved as an .mapak file

Ex: Load Custom Map Data

```
public void Rebuild(BinaryReader br)
{
    GameObject NewMap = Instantiate(TemplateObj, Space, 0);
    NewMap.name = "CustomMap" + index;
    index++;
    Parrents = NewMap.GetComponent<MapParrents>();

    byte[] bytes;
    int size = br.ReadInt32(); //got img size
    bytes = br.ReadBytes(size); //got img
    Parrents.SetImg(bytes);

    size = br.ReadInt32();
    for (int i = 0; i < size; i++)
    {
        Parrents.AddWalk(WalkPrefab, br);
    }

    size = br.ReadInt32();
    for (int i = 0; i < size; i++)
    {
        Parrents.AddExit(ExitPrefab, br);
    }

    size = br.ReadInt32();
    for (int i = 0; i < size; i++)
    {
        Parrents.AddBuilding(BuildPrefab, br);
    }

    NewMap.SetActive(false);
    CityList.AddCity(NewMap);
}
```

Img.

Walks

Exits

Buildings

Mod Order



CREATE



ADD



USE

1

Create Mod



2

Add Mod



3

Use Mod



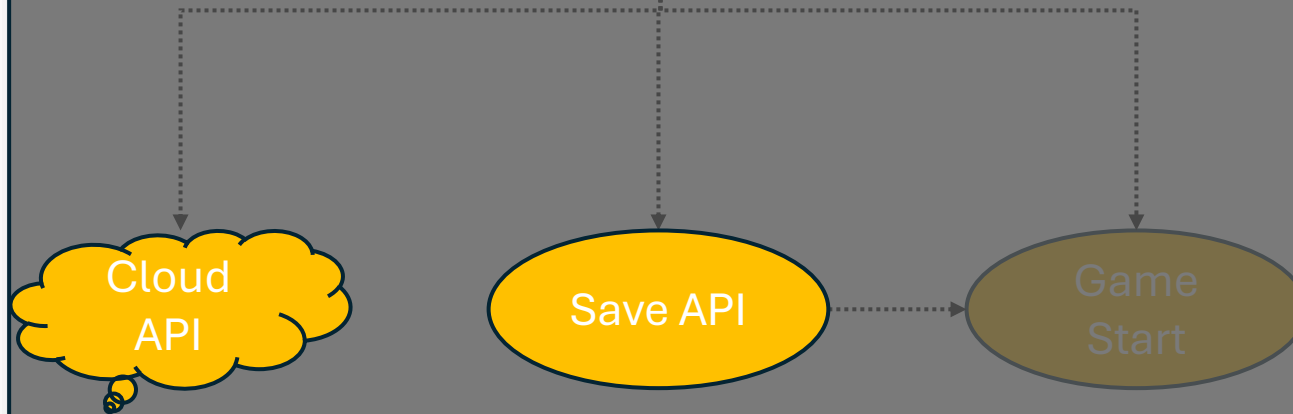
Mod Item Data

Mod Desc

Name
Keyword
Version Number
Creator
Description
Photo

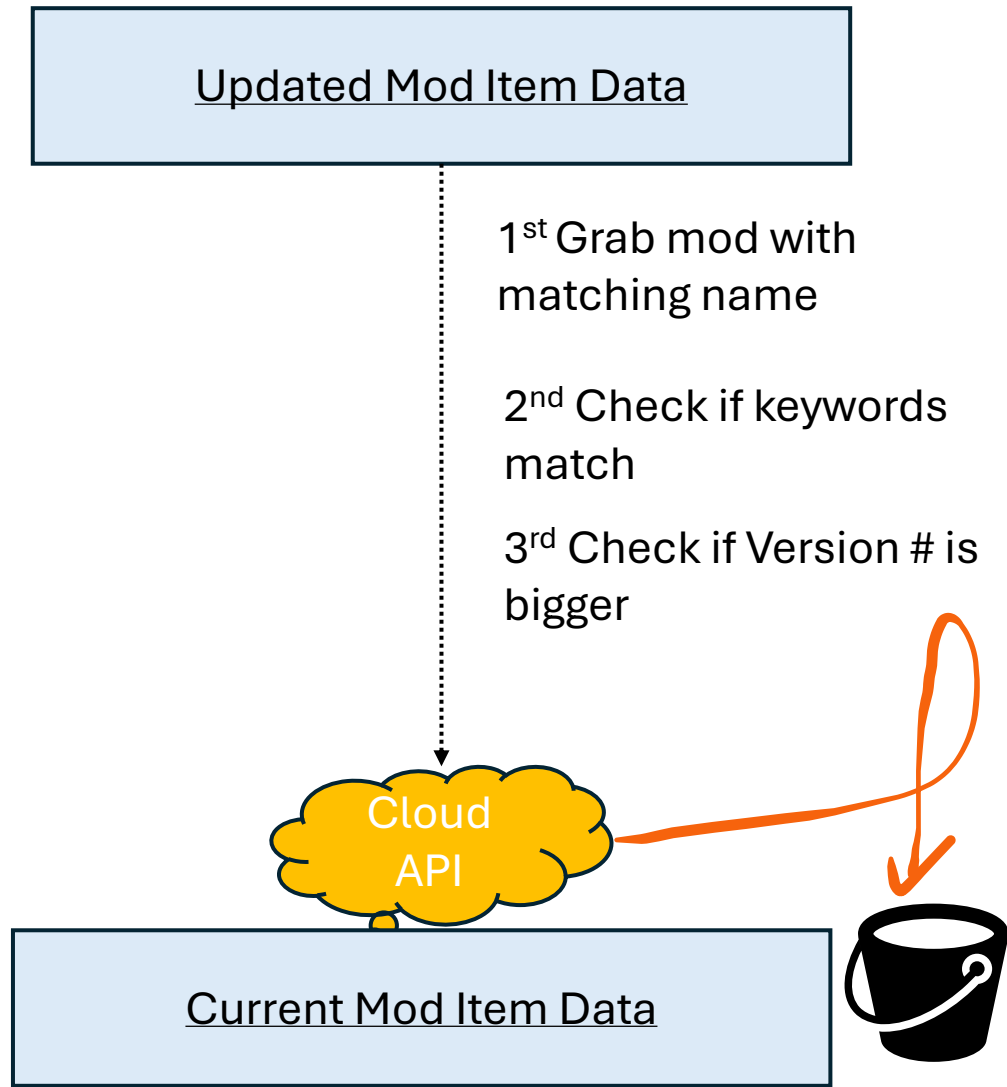
Mod Contents

Mode Type
Mod Data



Update Mods

Requires Keyword and a higher version number.



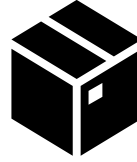
Change in Cloud

If all checks pass, the cloud Api will replace with new mod data

1st Check if Mod needs
an update

Save API

2nd Grab Mod



3rd Grab Cloud Mod

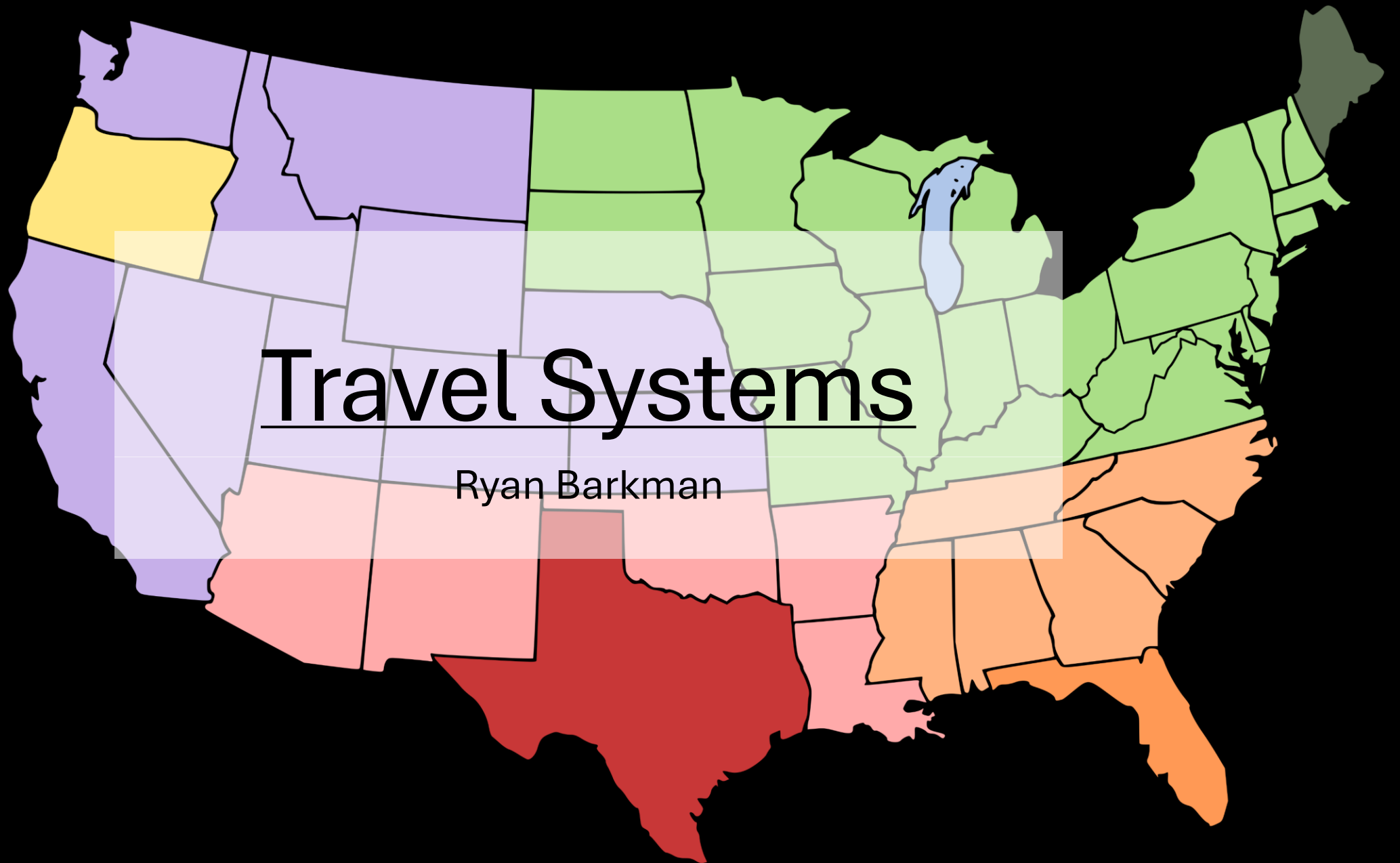
4th If version # is higher:
Replace In storage

Cloud
API



Update In Game

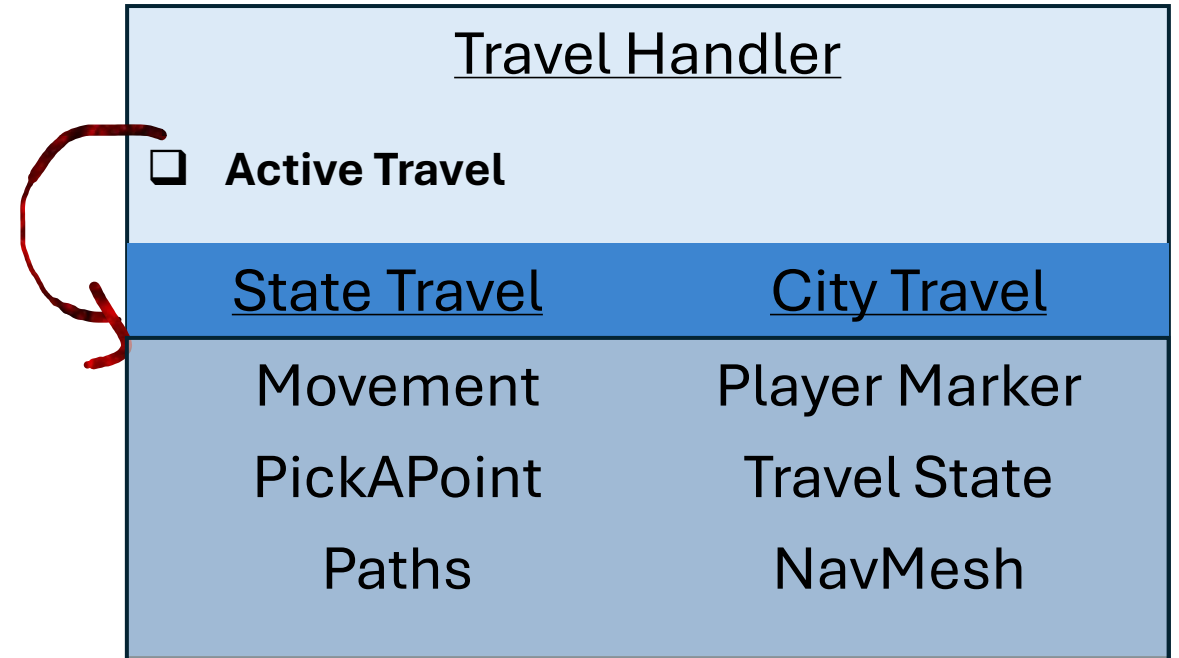
Requires Keyword and a higher
version number.



Travel Systems

Ryan Barkman

Travel Handlers



Player Input

Player clicks on map

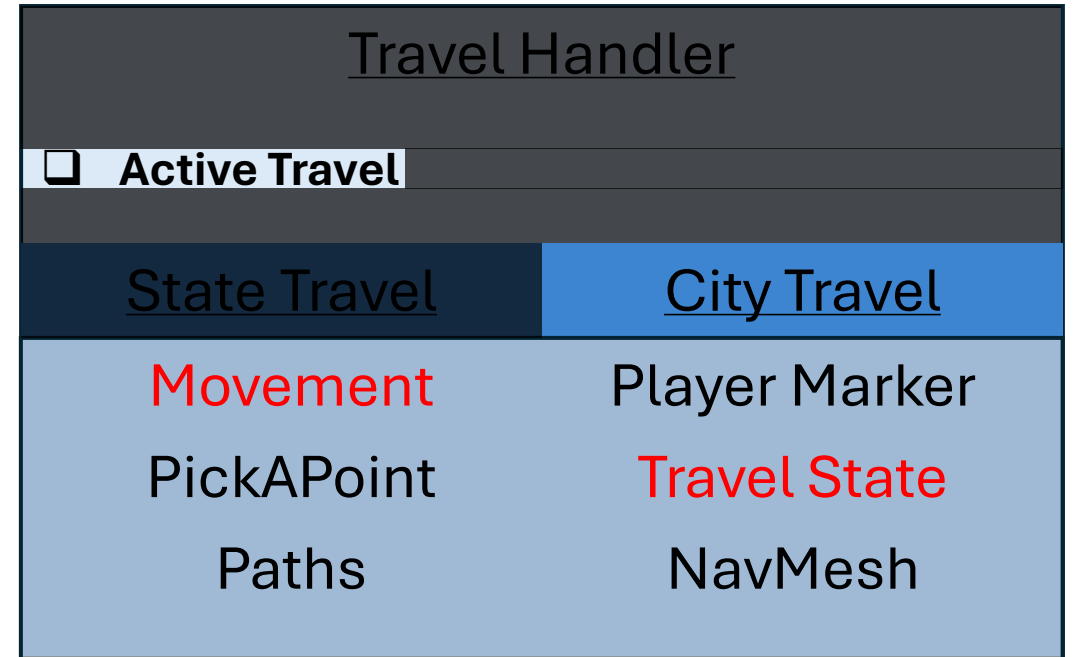
System sends data to the active travel node

Travel Handler	
☐ Active Travel	
State Travel	City Travel
Movement	Player Marker
PickAPoint	Travel State
Paths	NavMesh

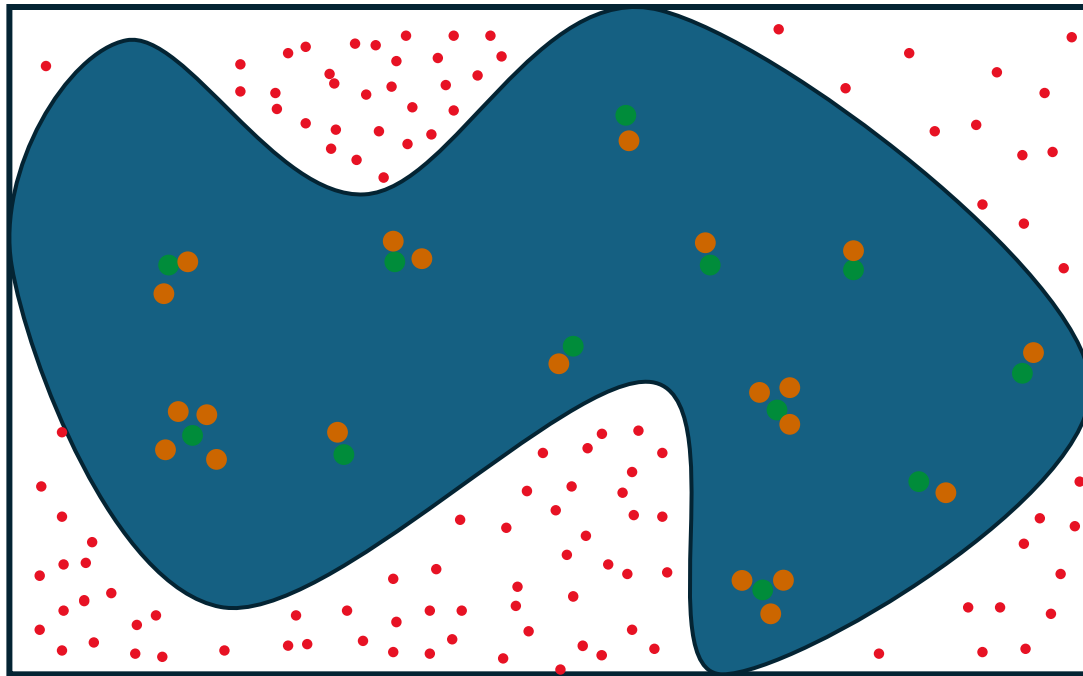
Game Update

Travel Handler has its own
update method

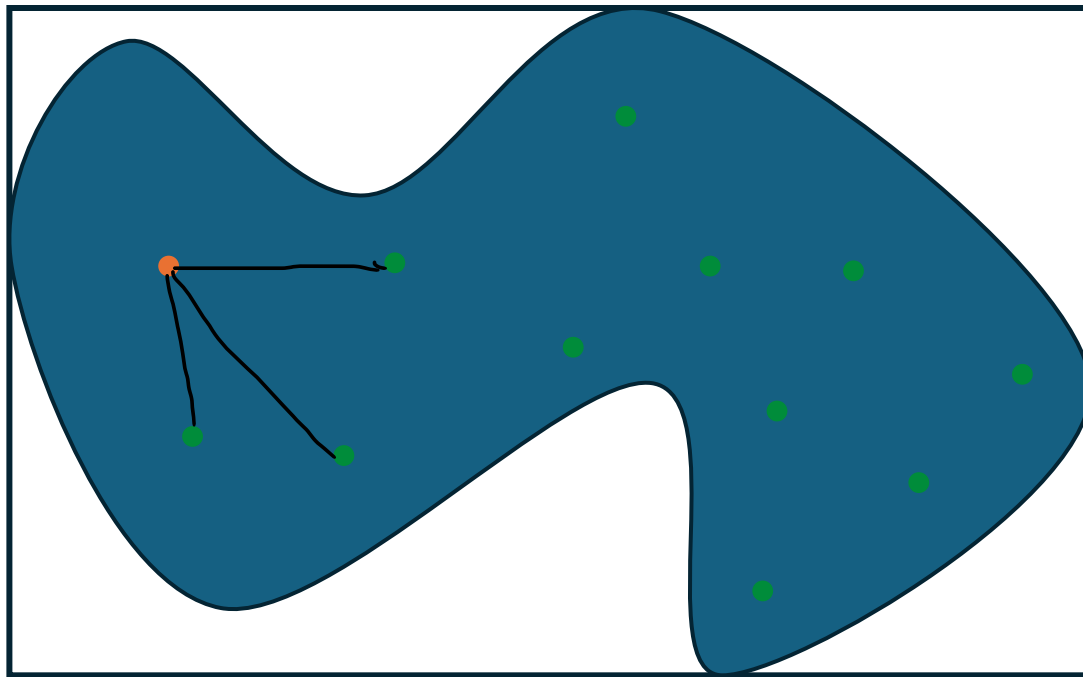
Ensures only Active travel node
gets updated



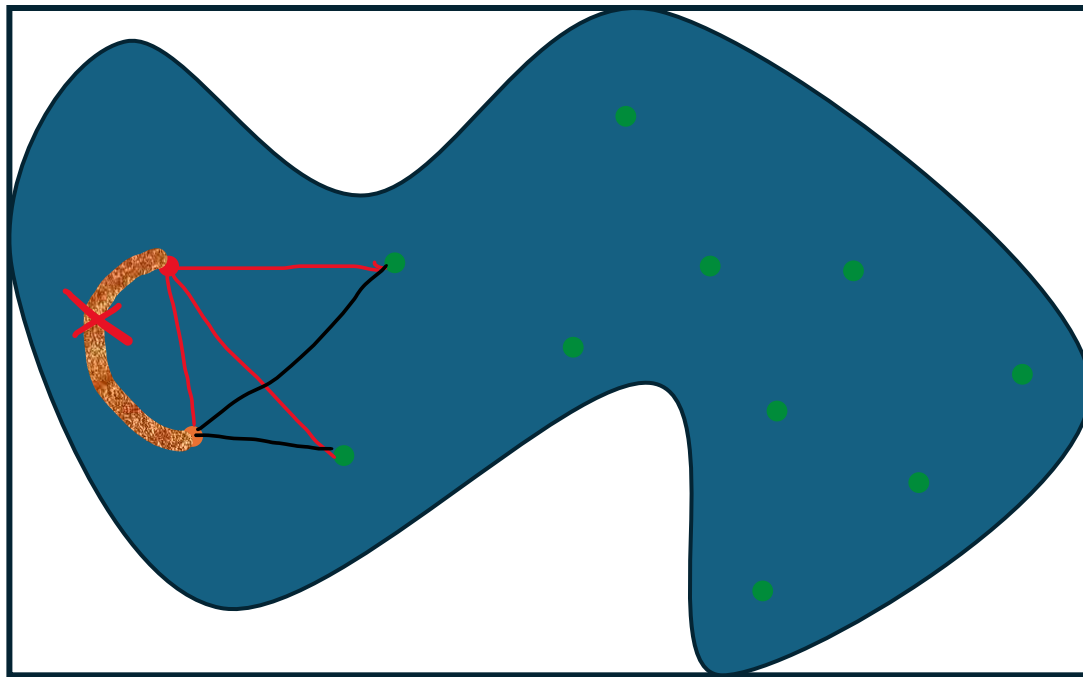
City Spawning



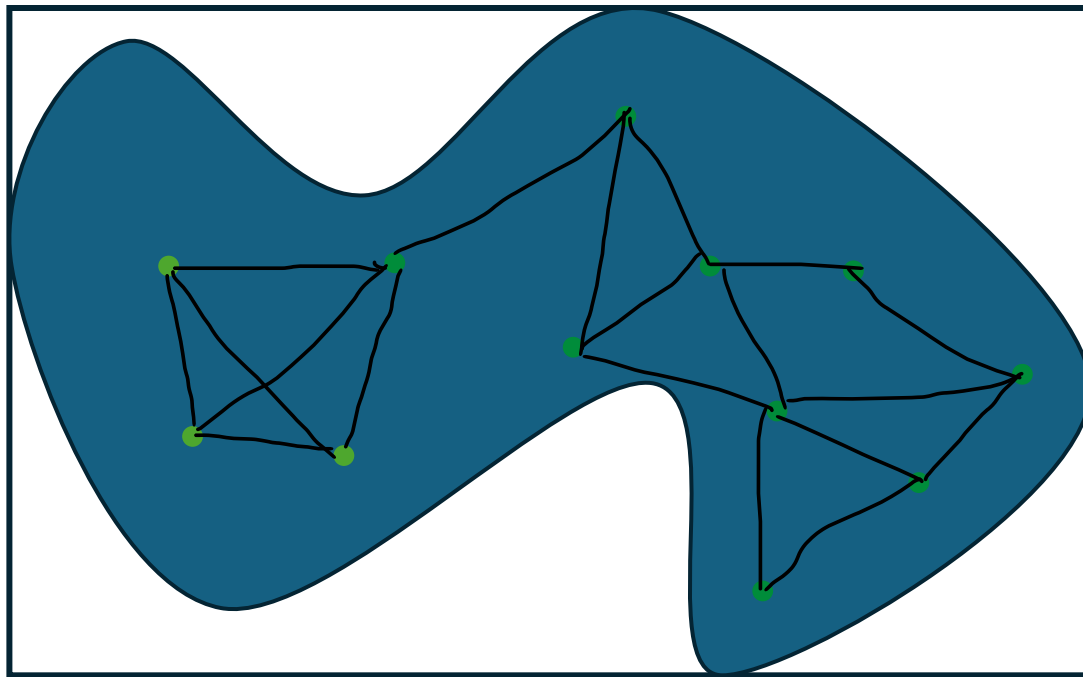
Pathways Spawning

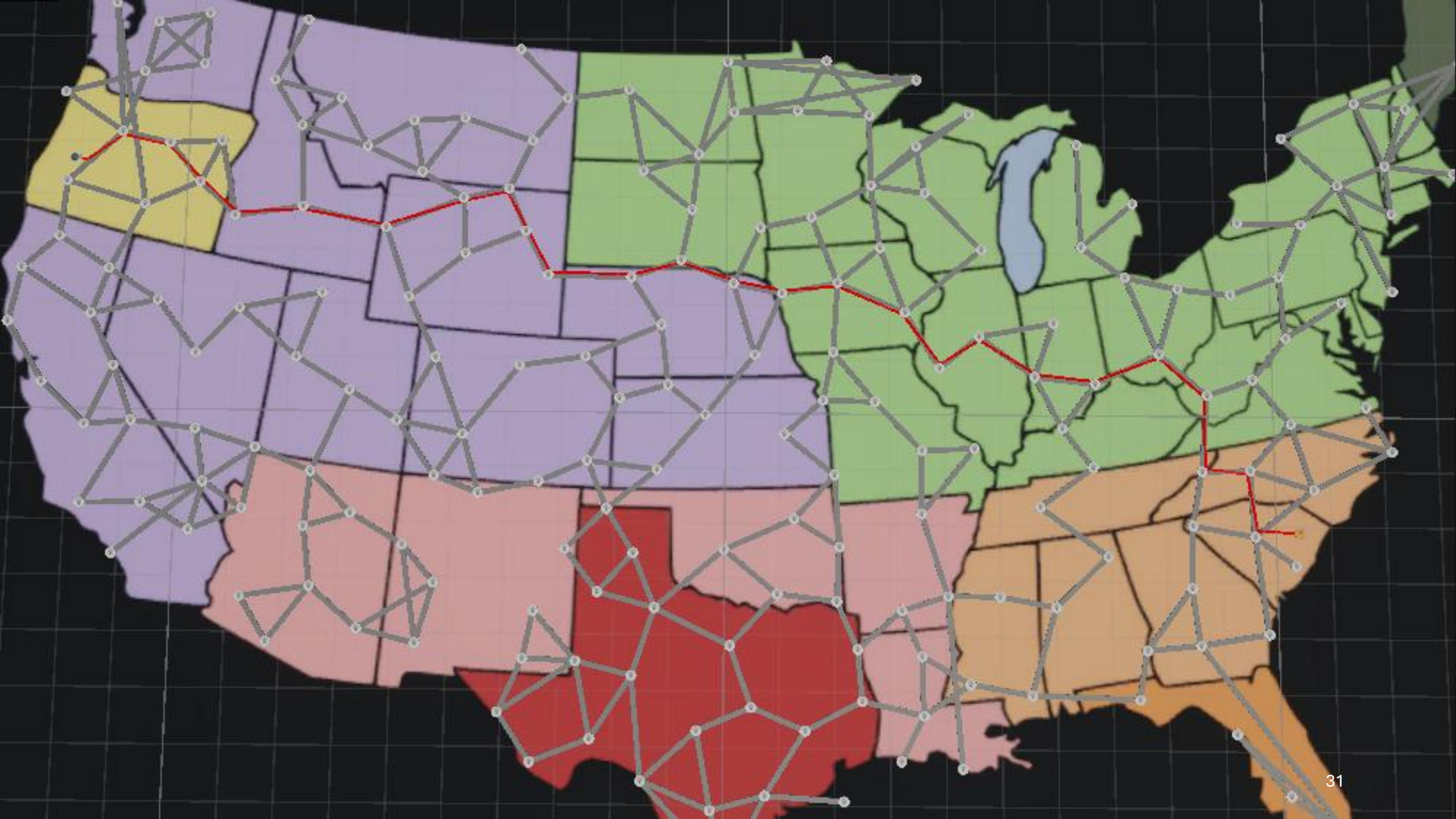


Pathways Spawning



Example





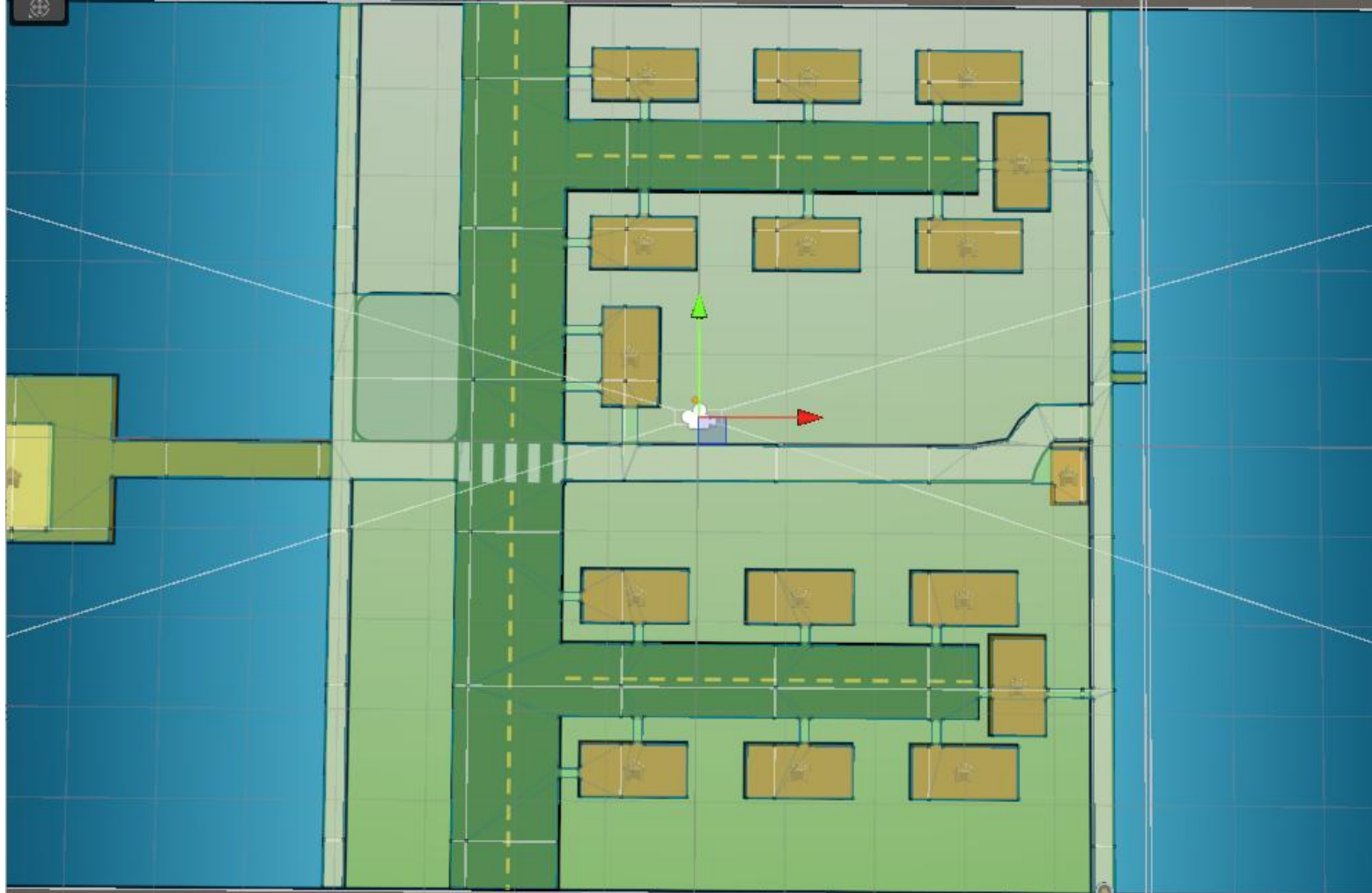
Nav-Mesh

```
NavMesh.CalculatePath(Start, End, Area, Path);
```

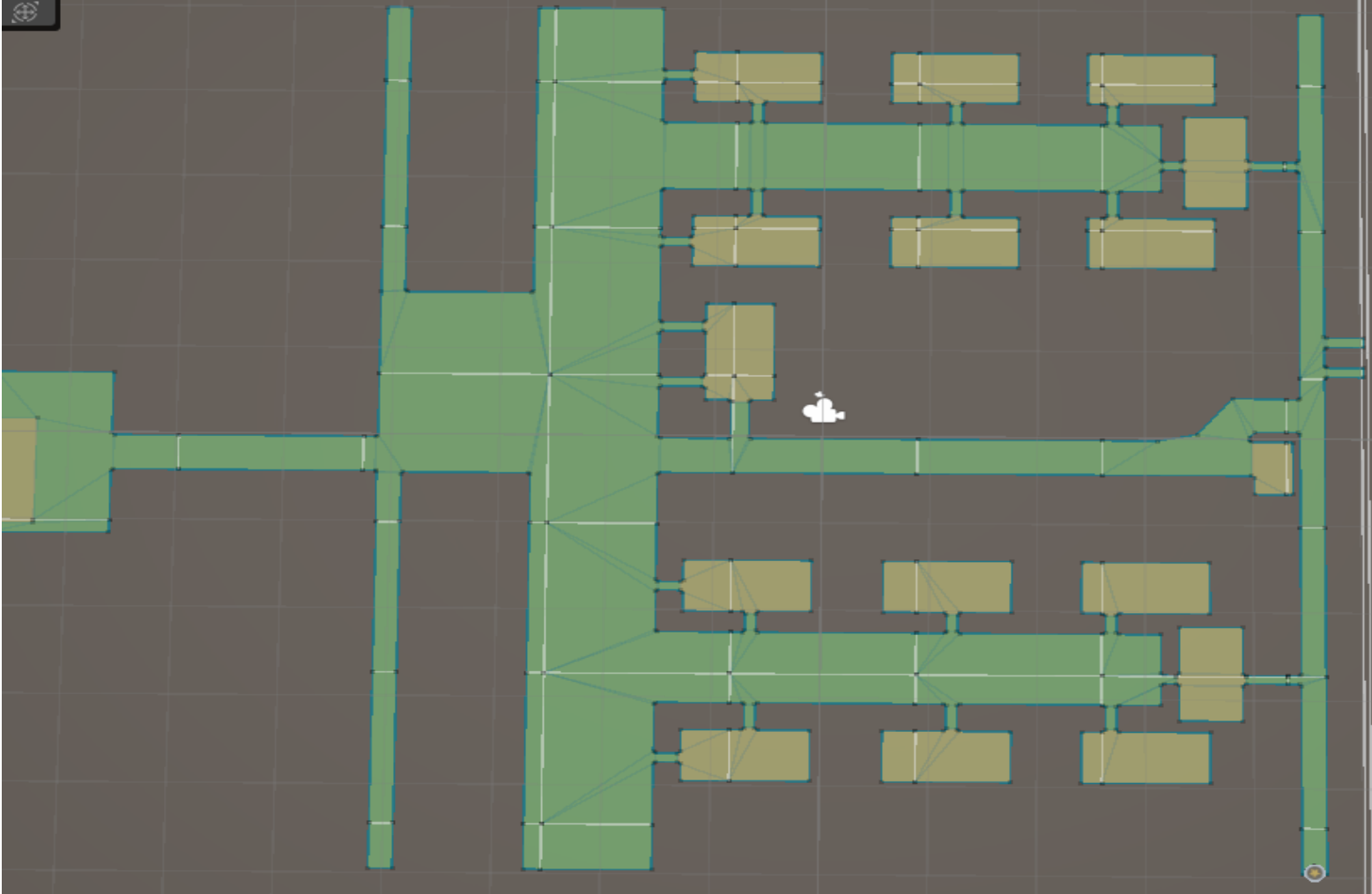
Navigation		
Agents Areas		
	Name	Cost
 Built-in 0	Walkable	1
 Built-in 1	Not Walkable	1
 Built-in 2	Jump	2
 User 3	Pathway	5
 User 4	Map	10
 User 5	Nope	0
 User 6	Building	36

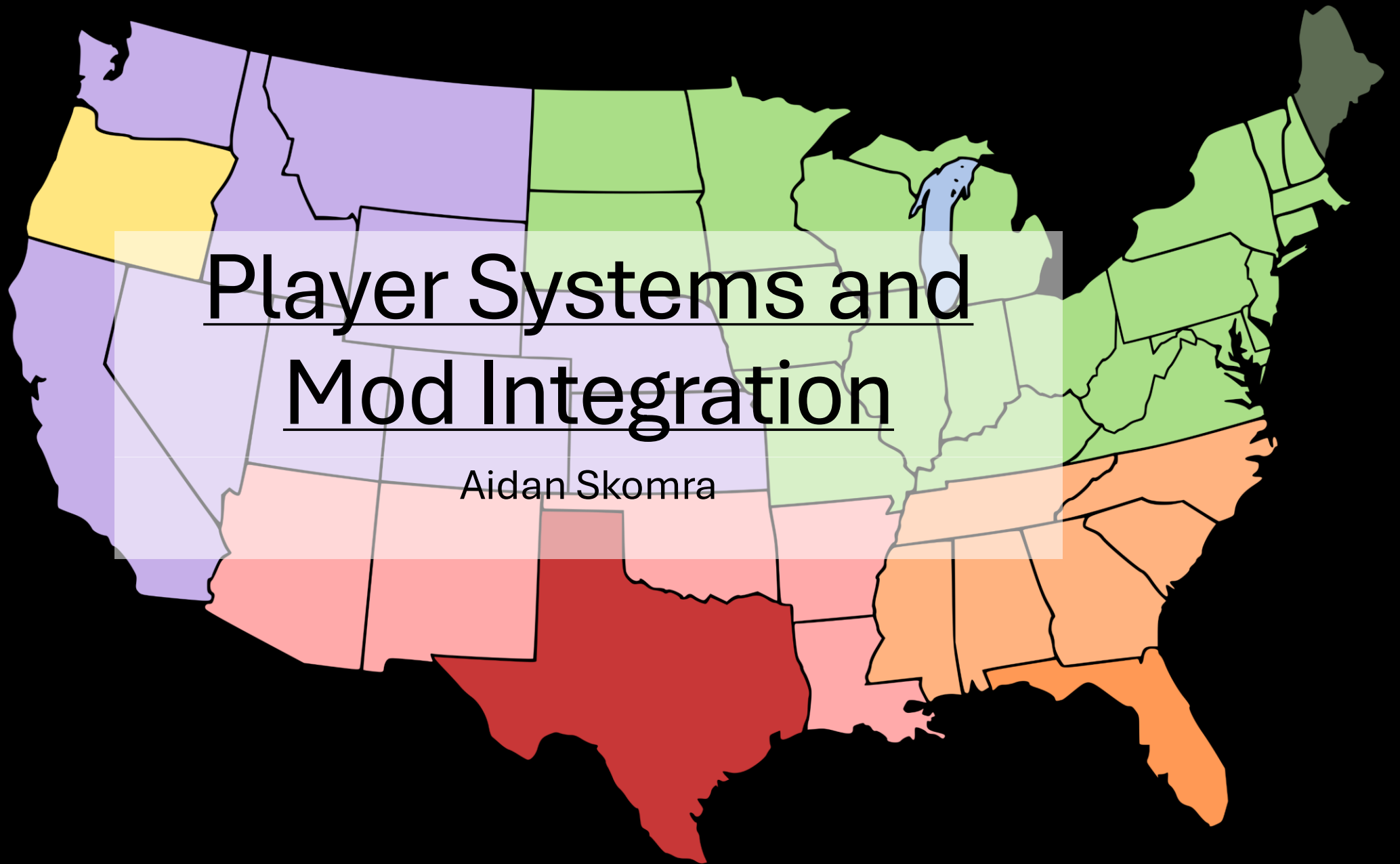
Use Geometry	Physics Colliders
▼ Object Collection	
Collect Objects	All Game Objects
Include Layers	Location, Pathway

Nav-Mesh



Nav-Mesh

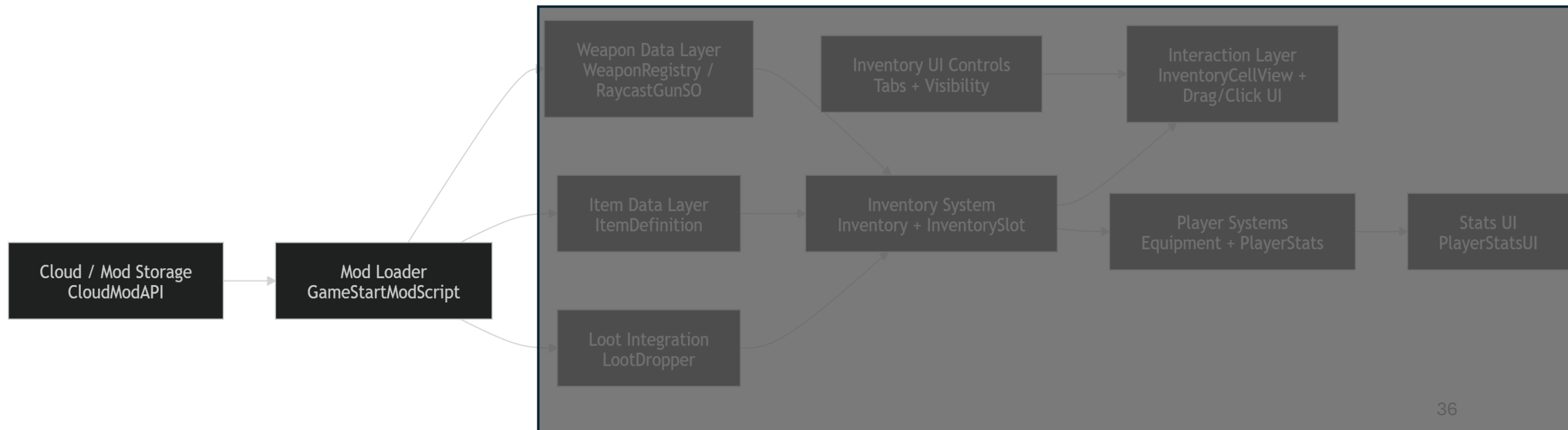
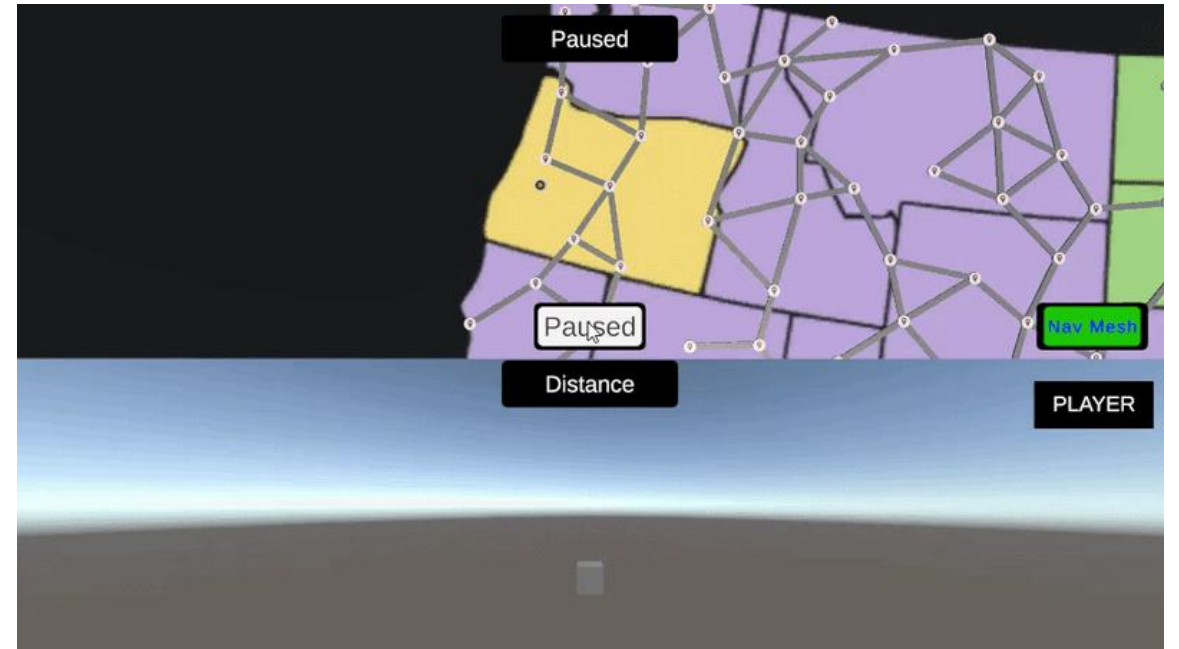




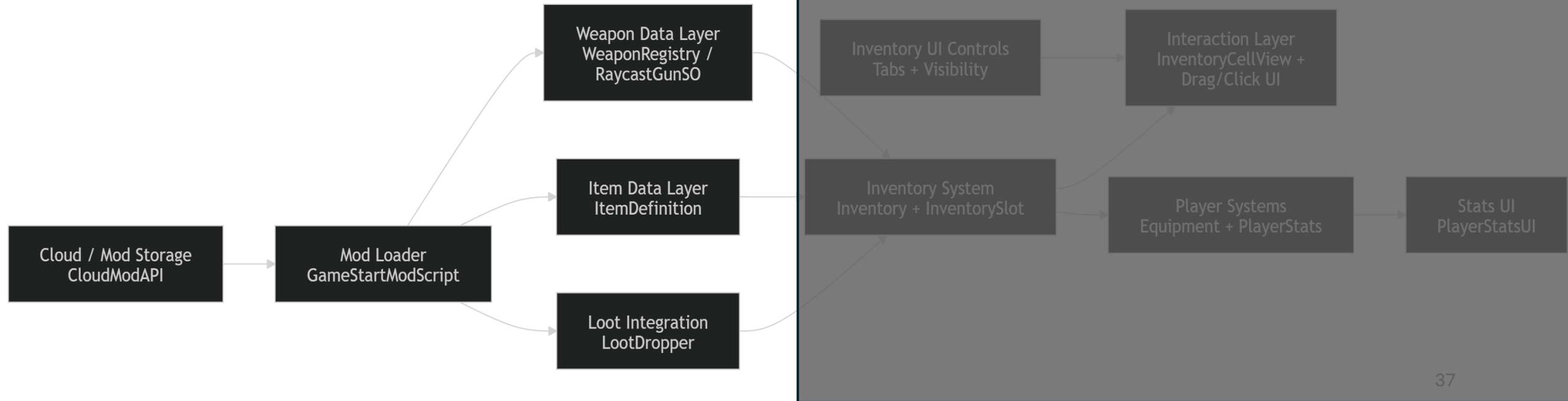
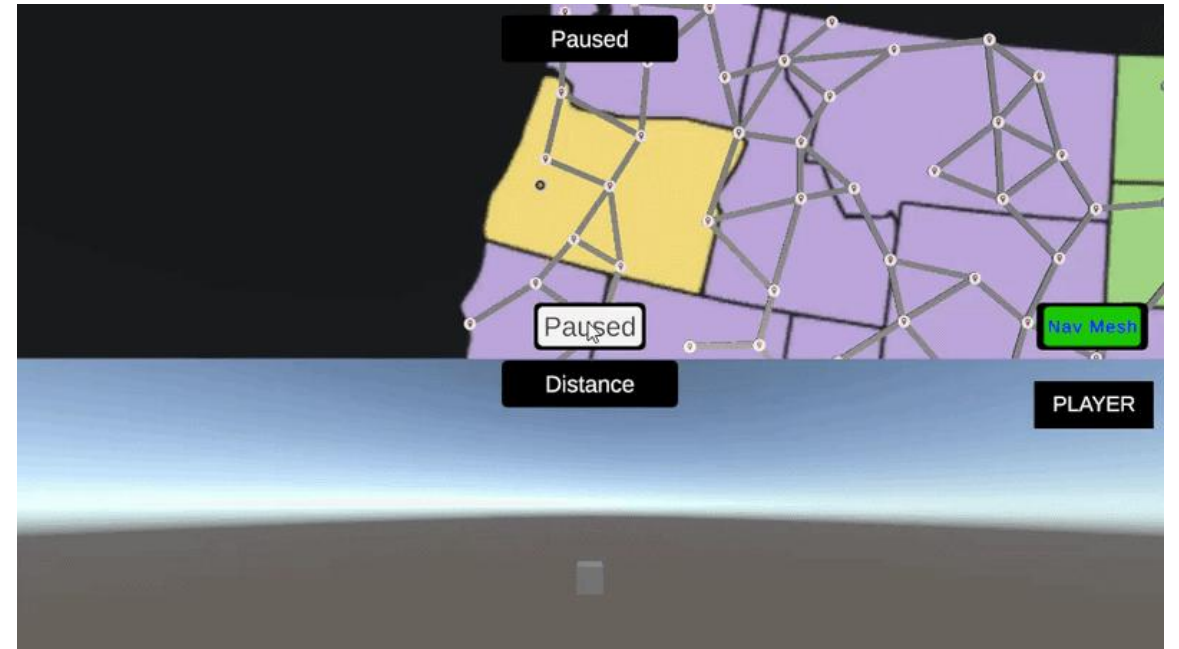
Player Systems and Mod Integration

Aidan Skomra

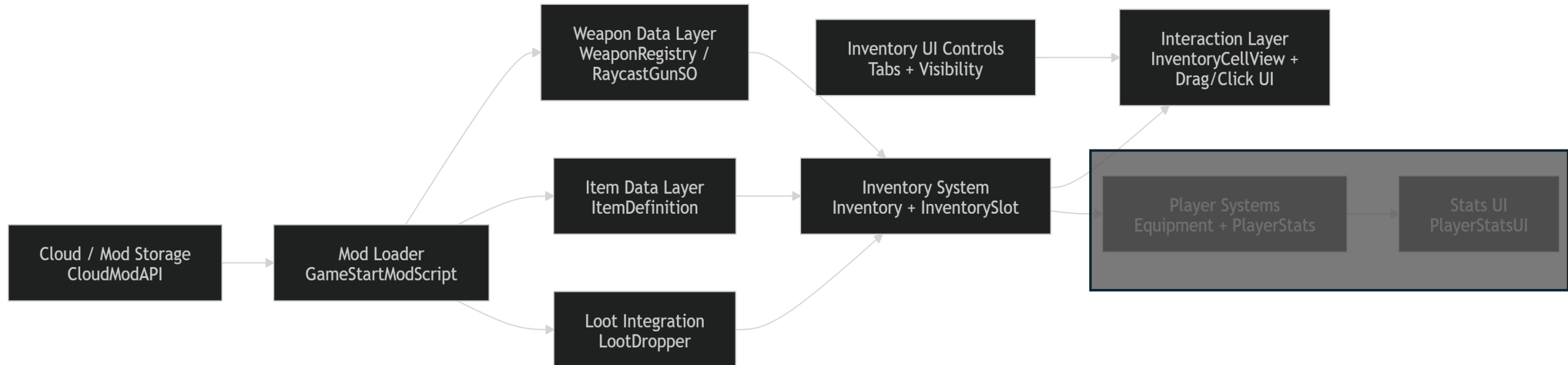
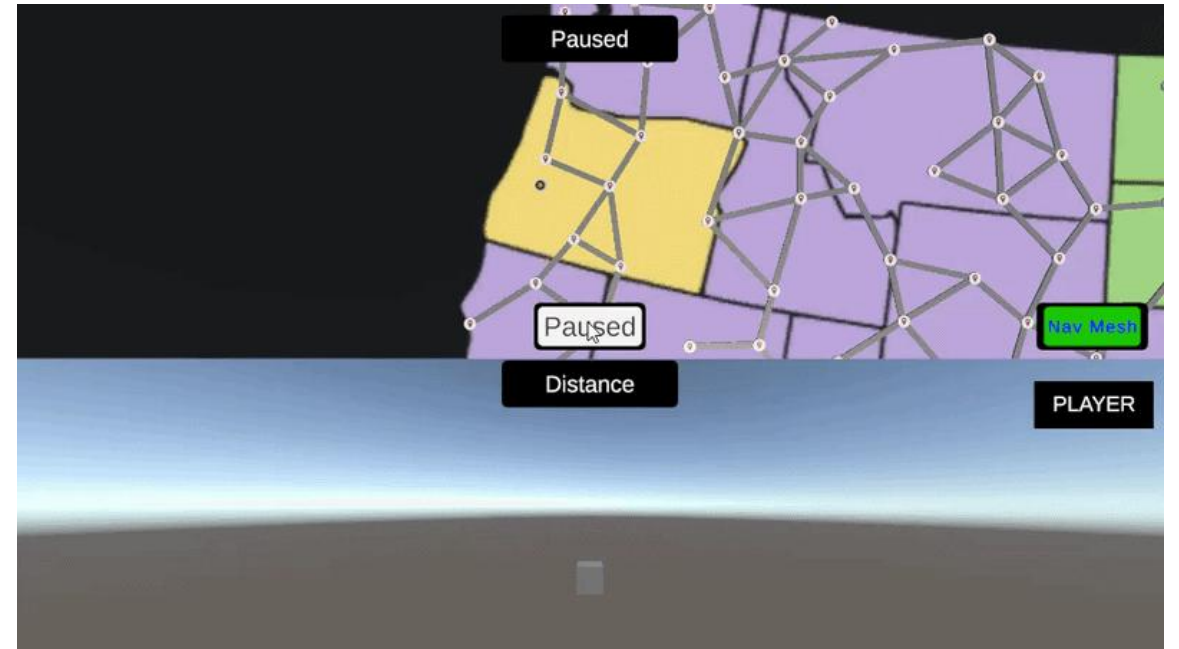
System Overview



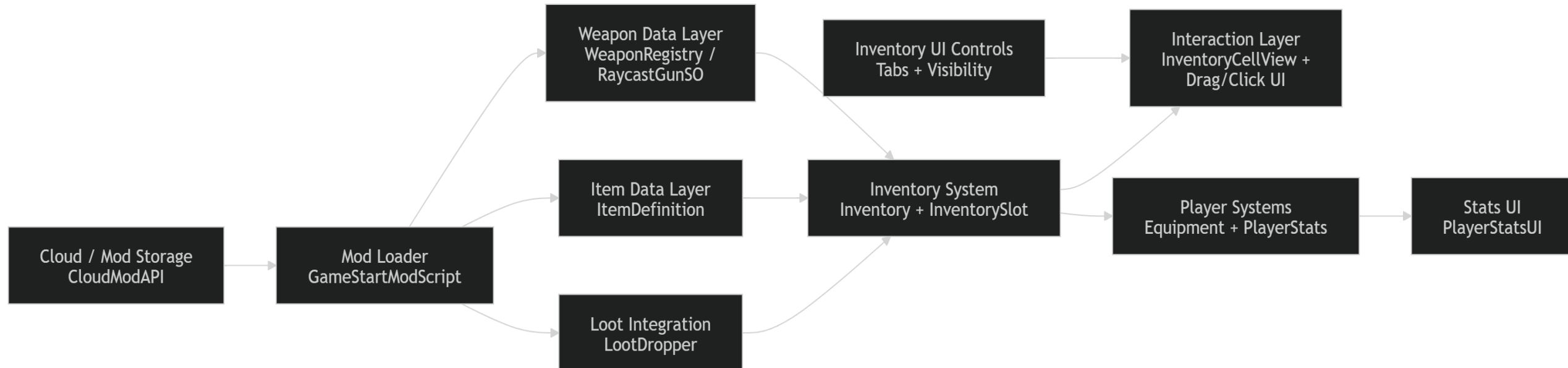
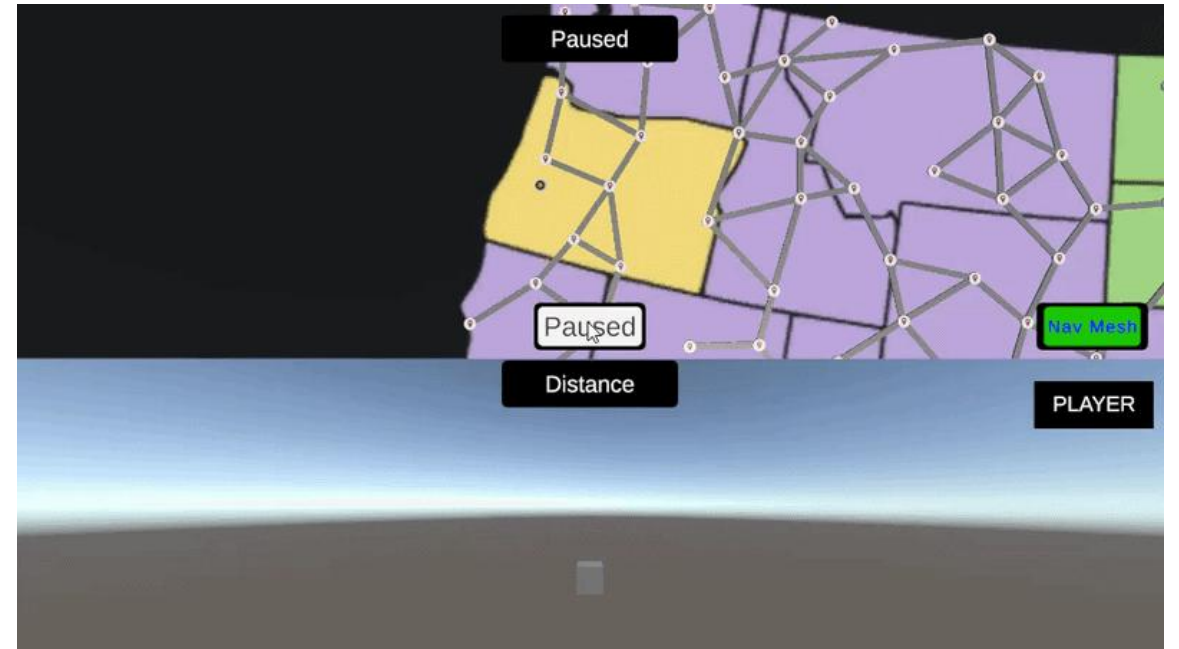
System Overview



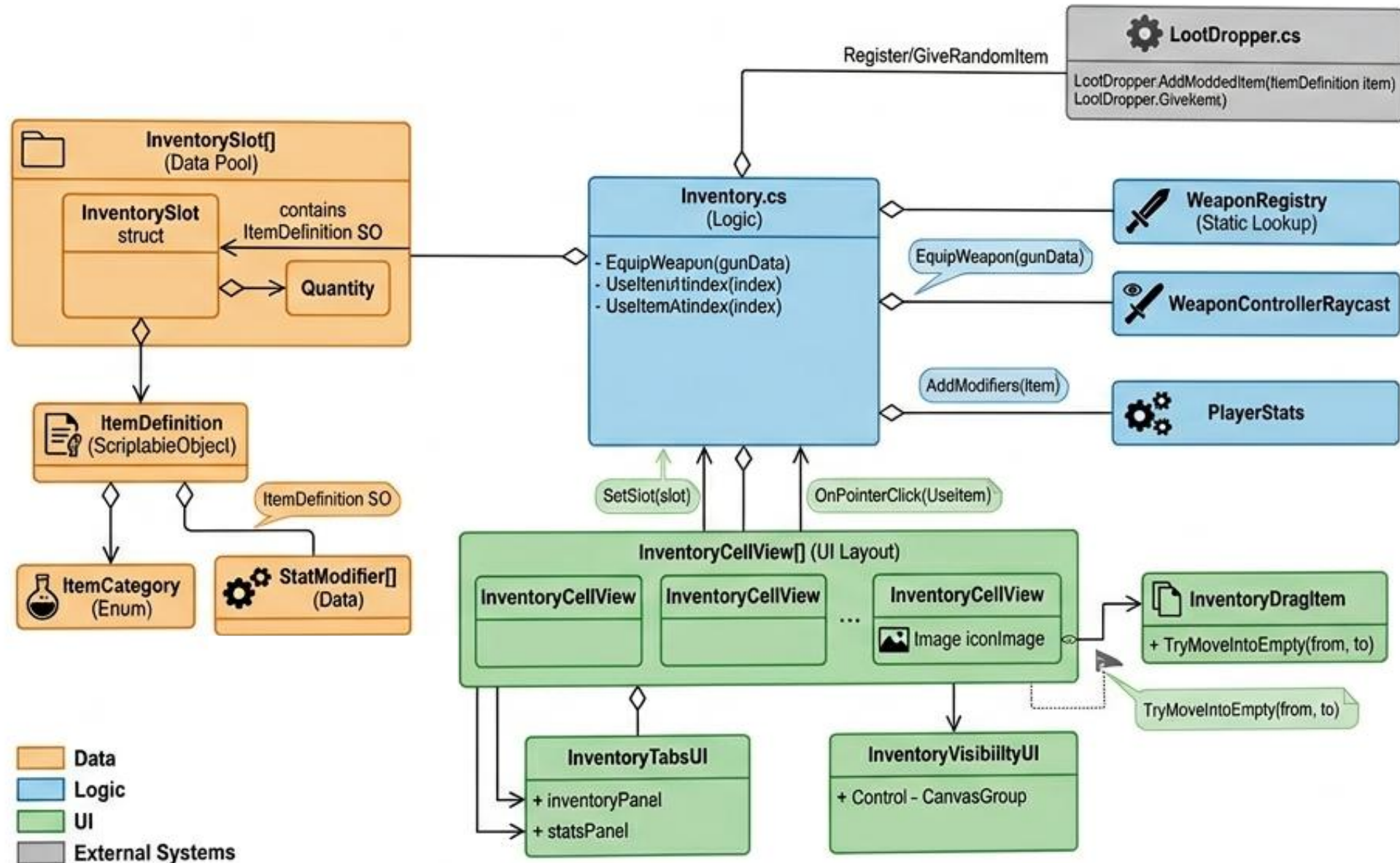
System Overview



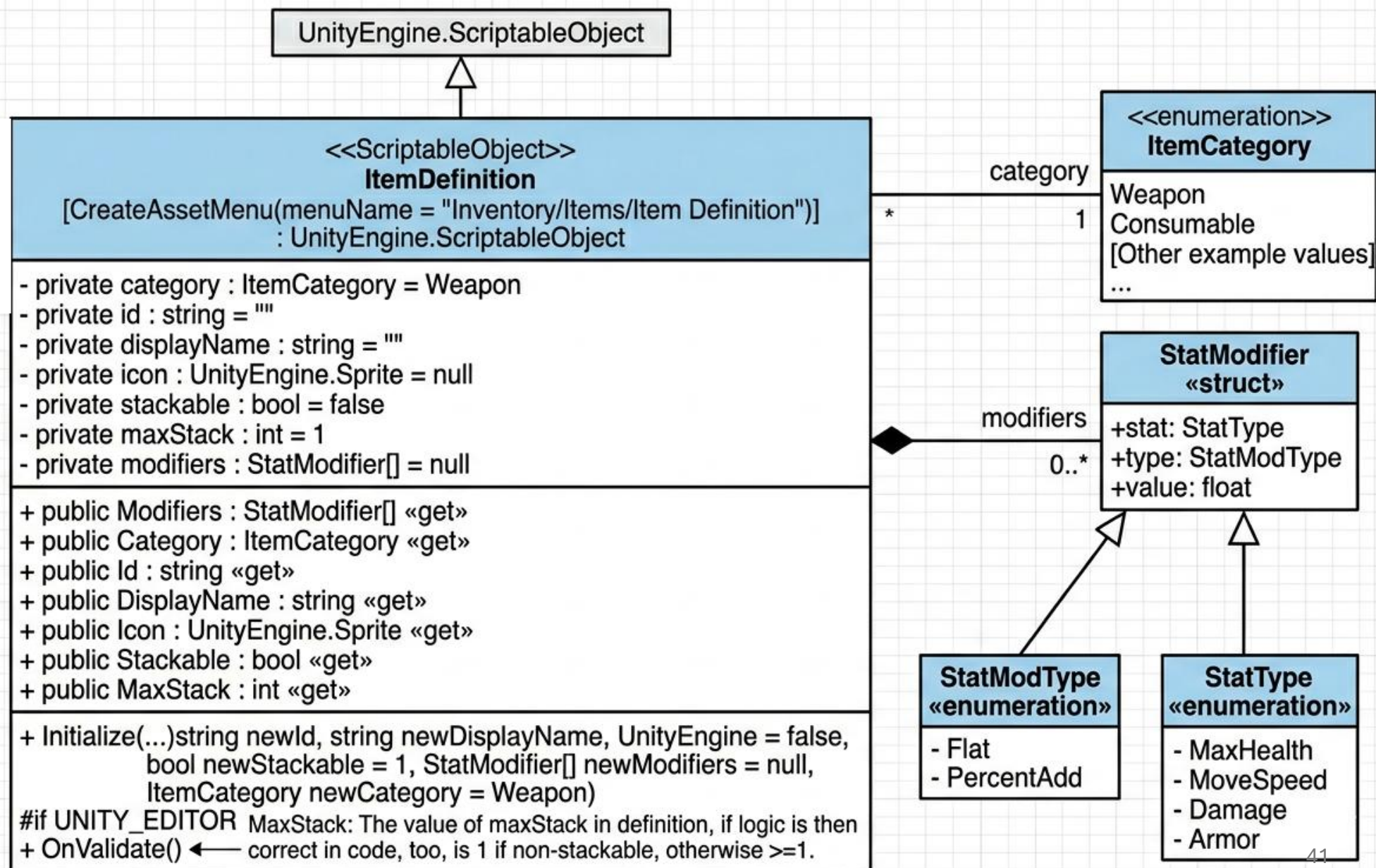
System Overview



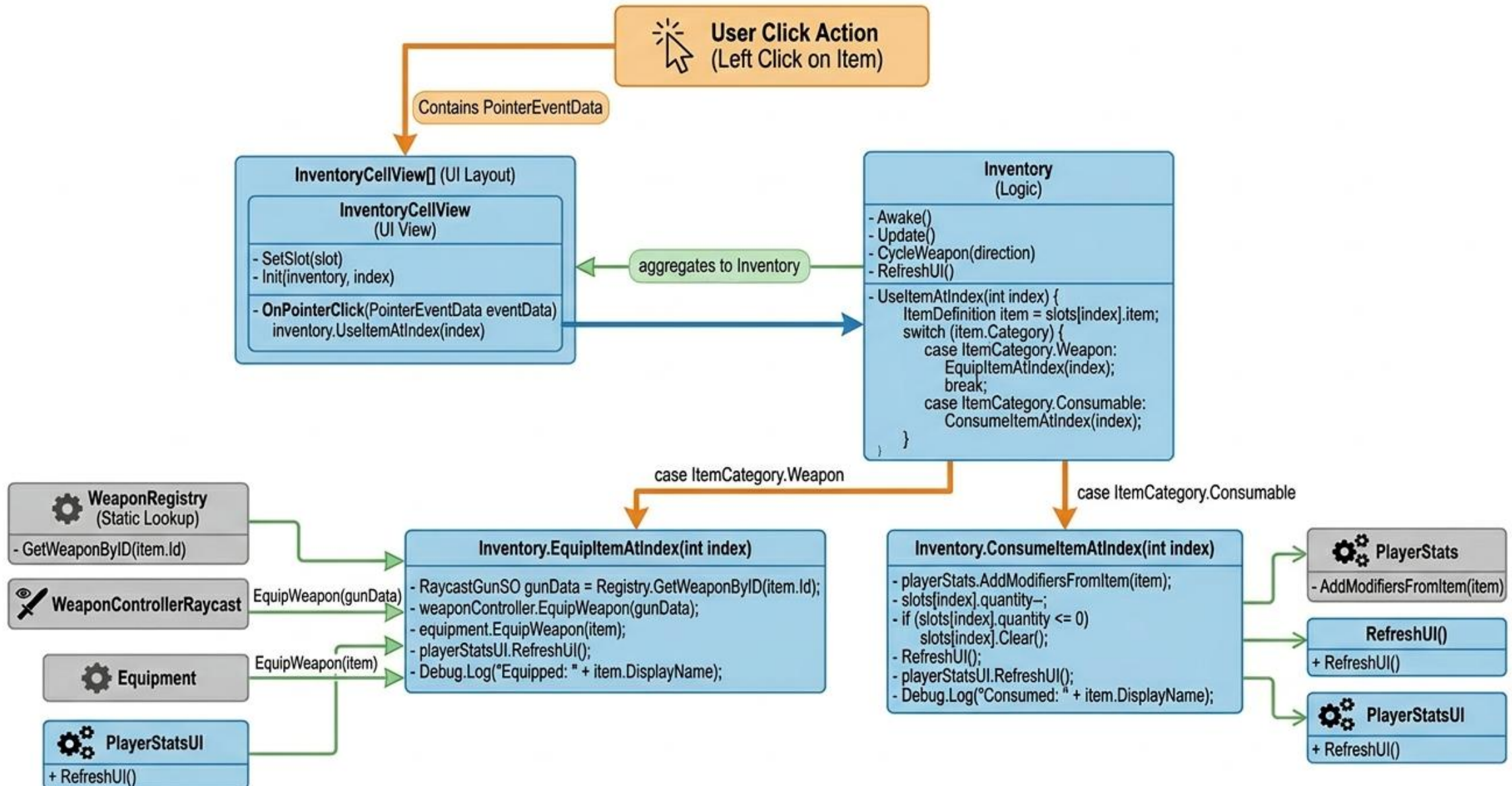
Inventory System Overview



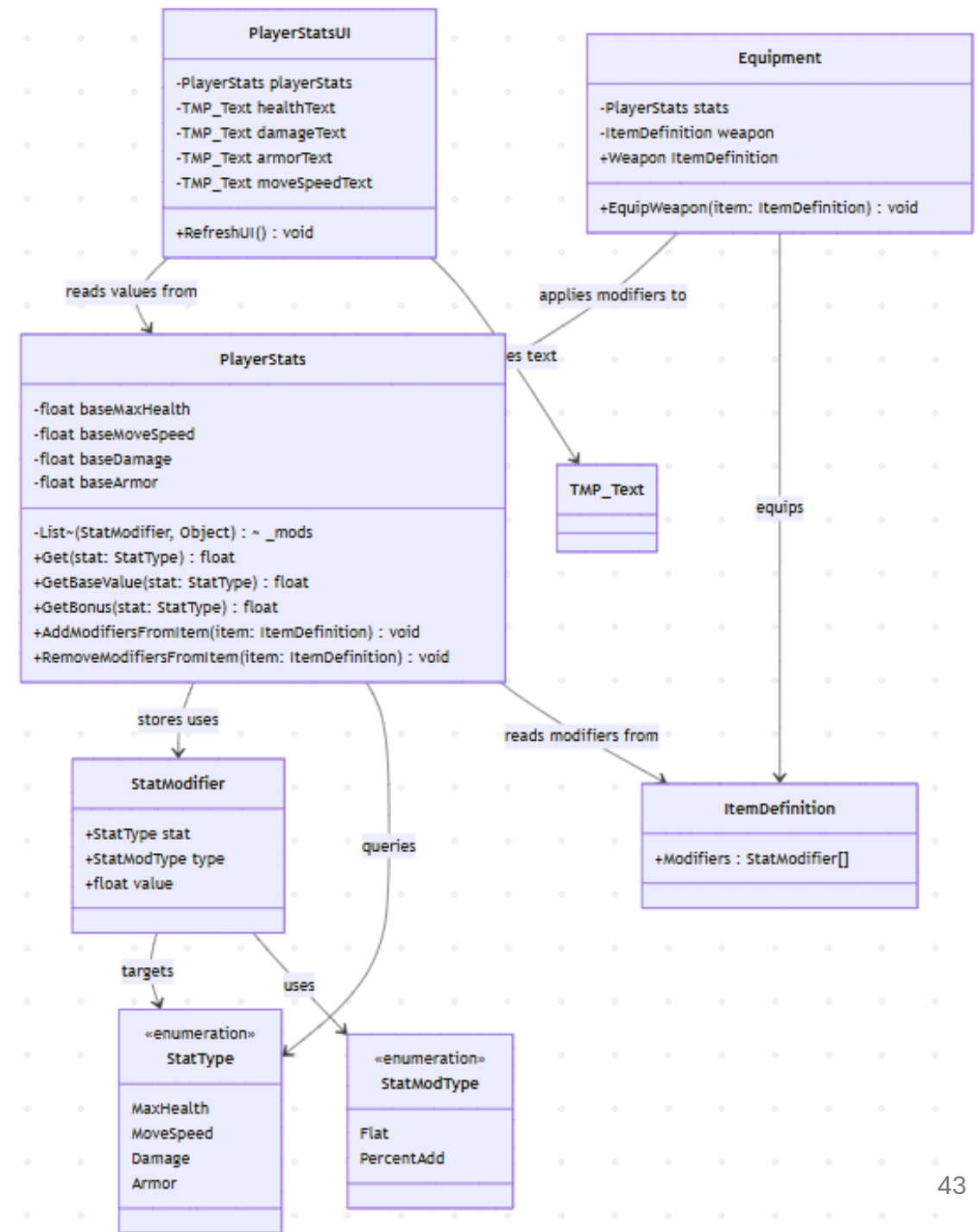
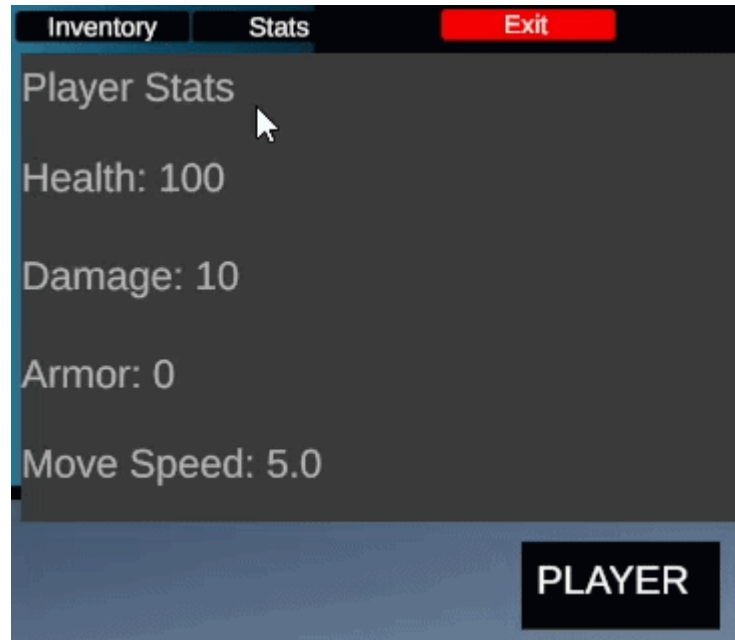
Item System (Item Definition)



Item Interaction Model

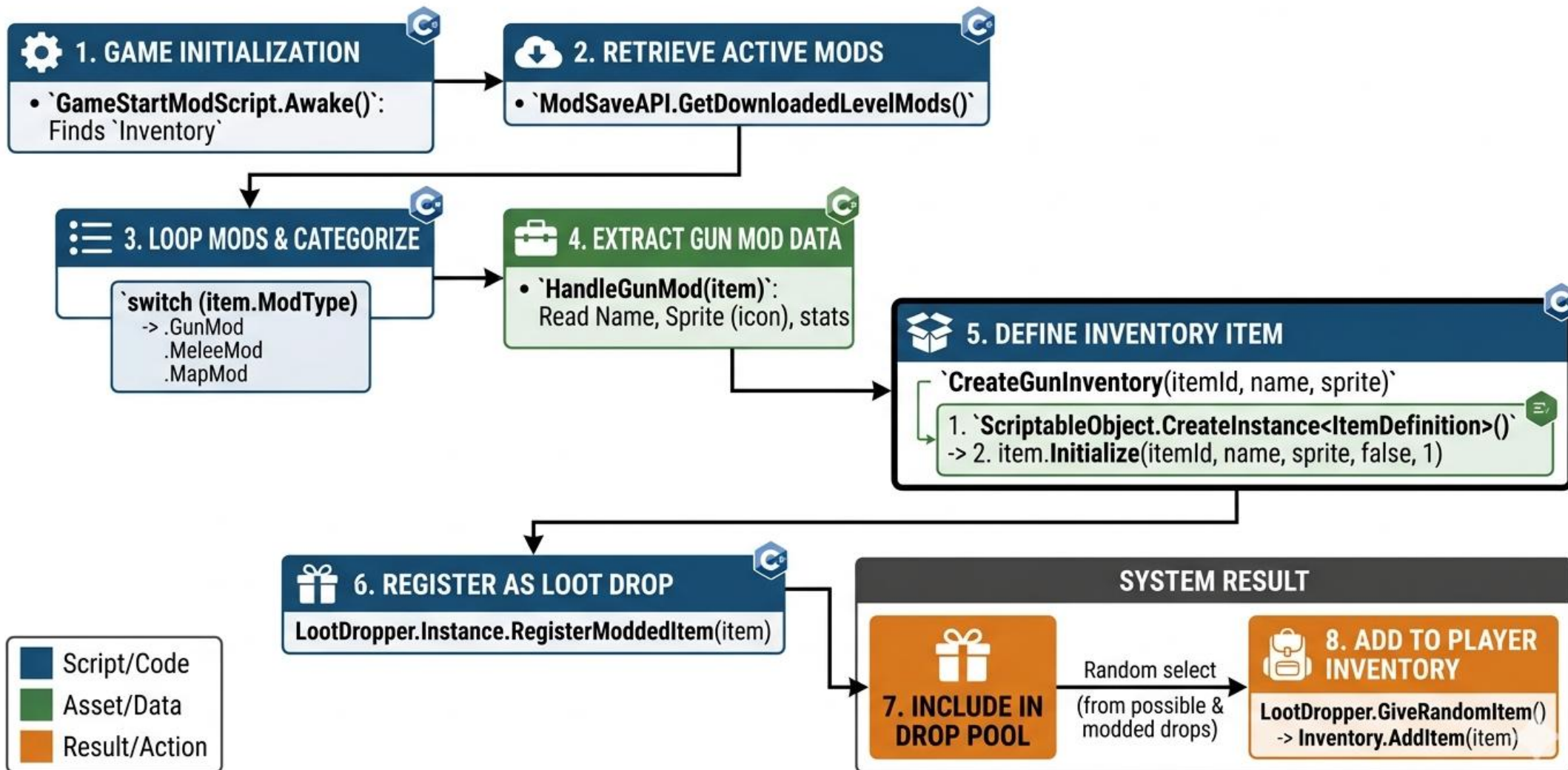


Player Stats System

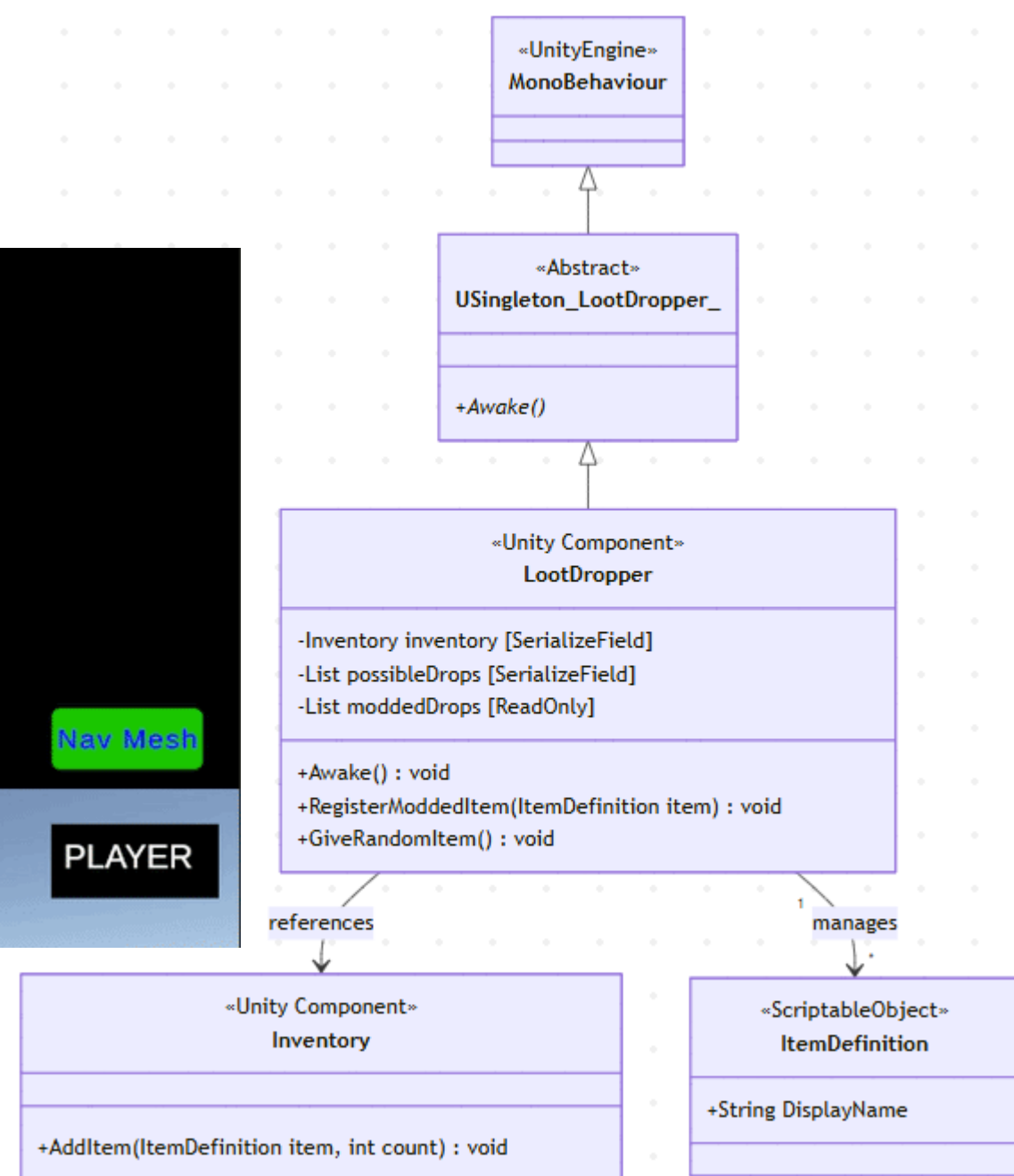
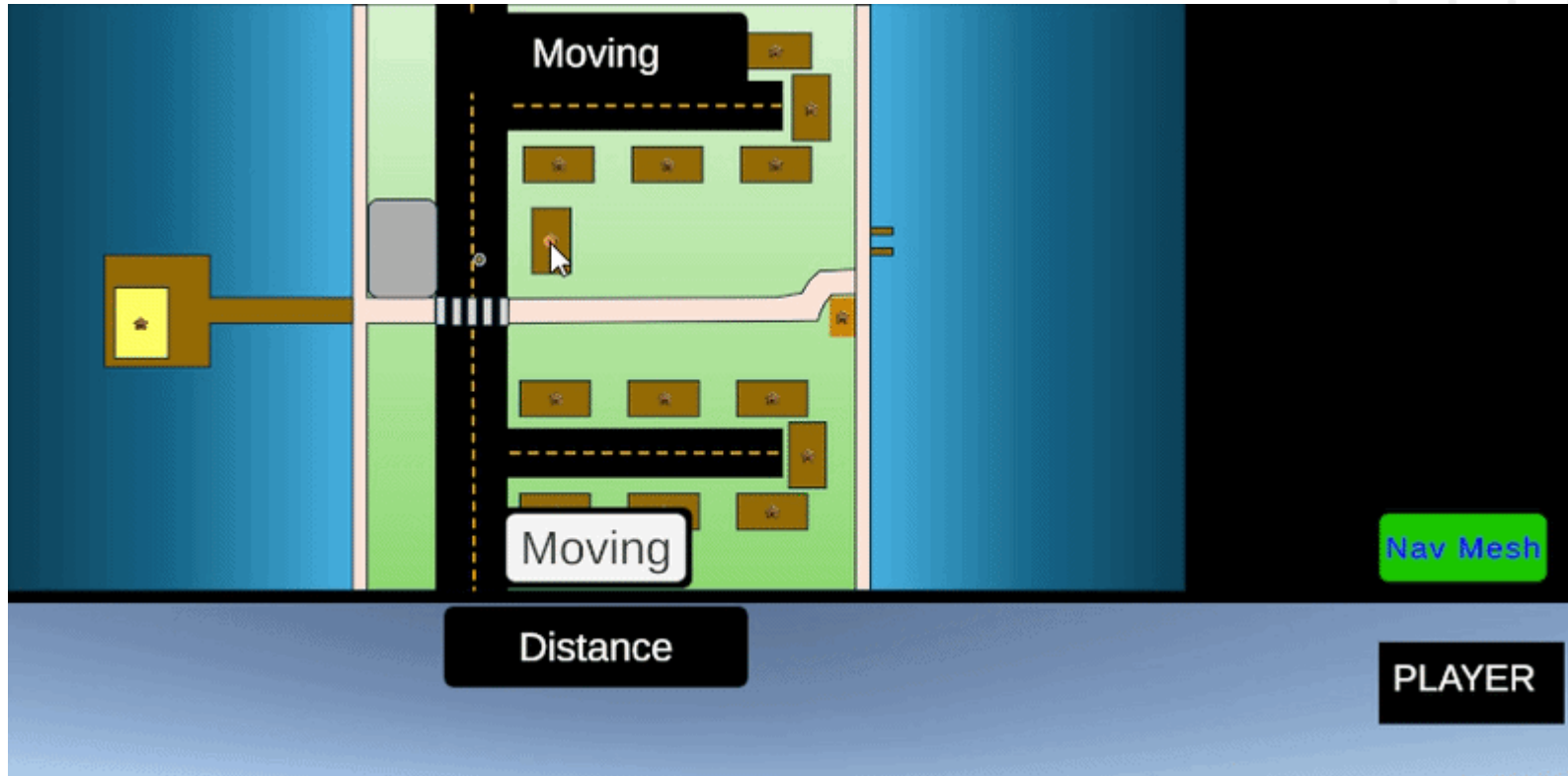


Mod System

GAMESTART MODDING FLOW: INVENTORY INTEGRATION

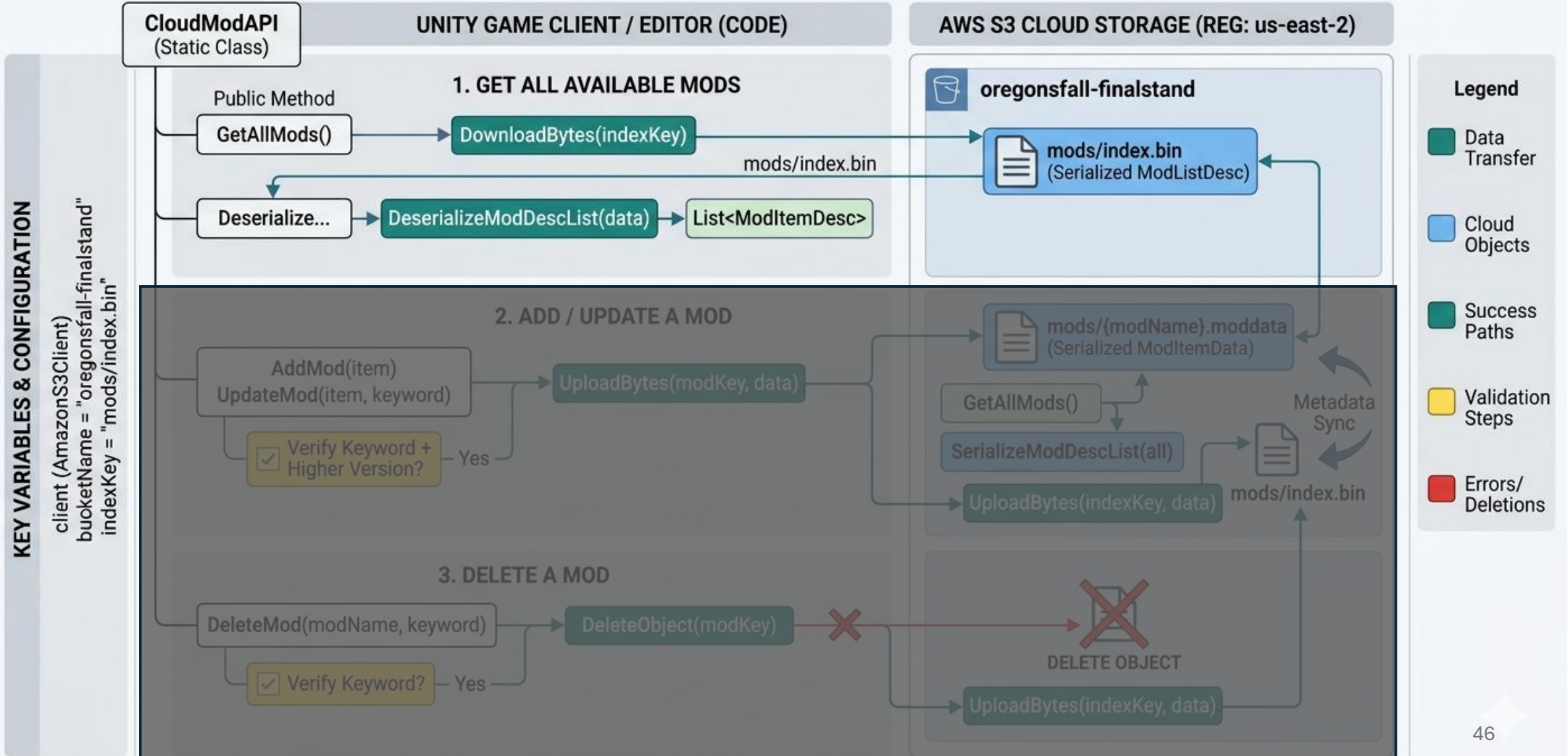


Looting System



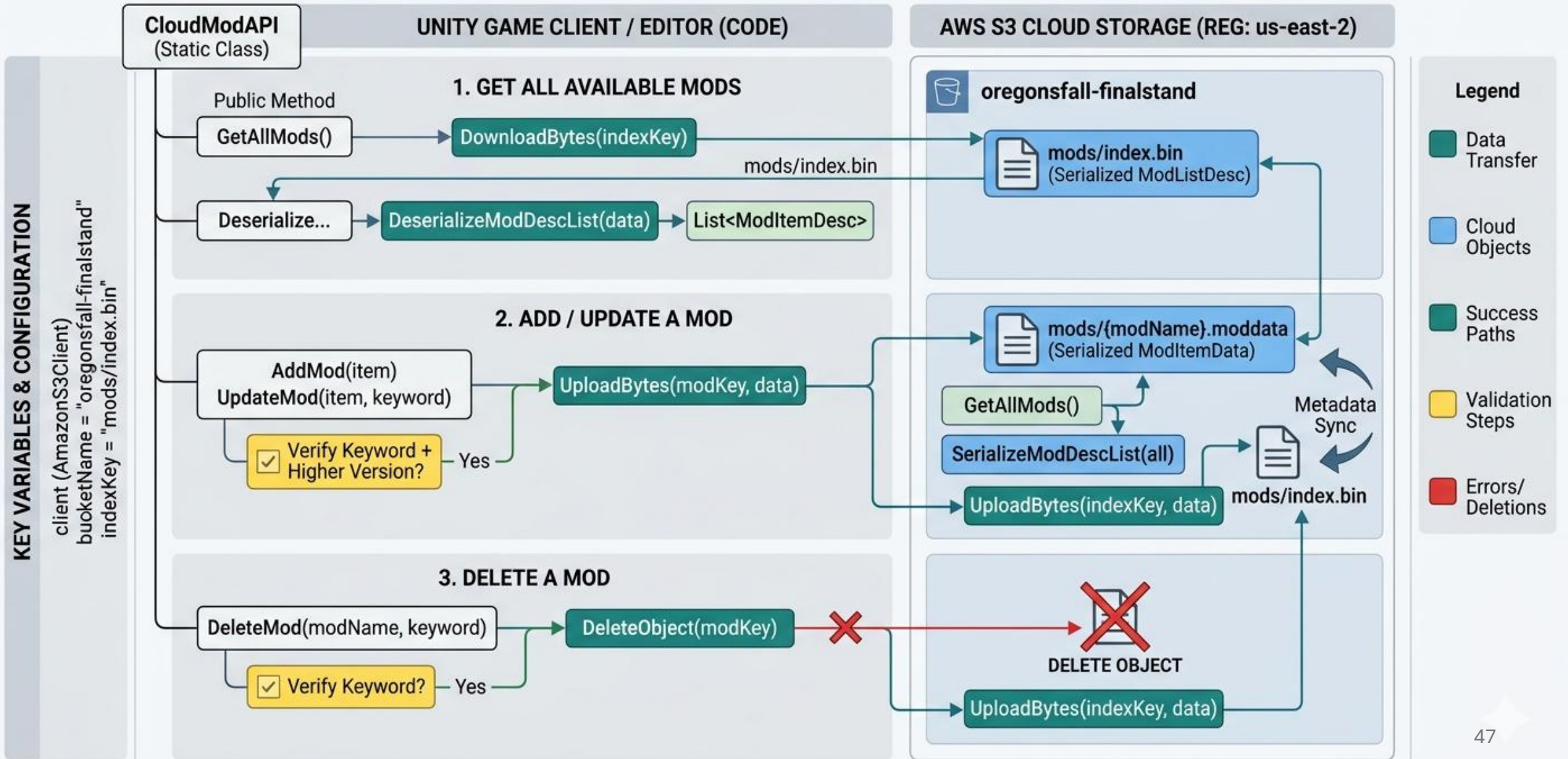
Cloud System

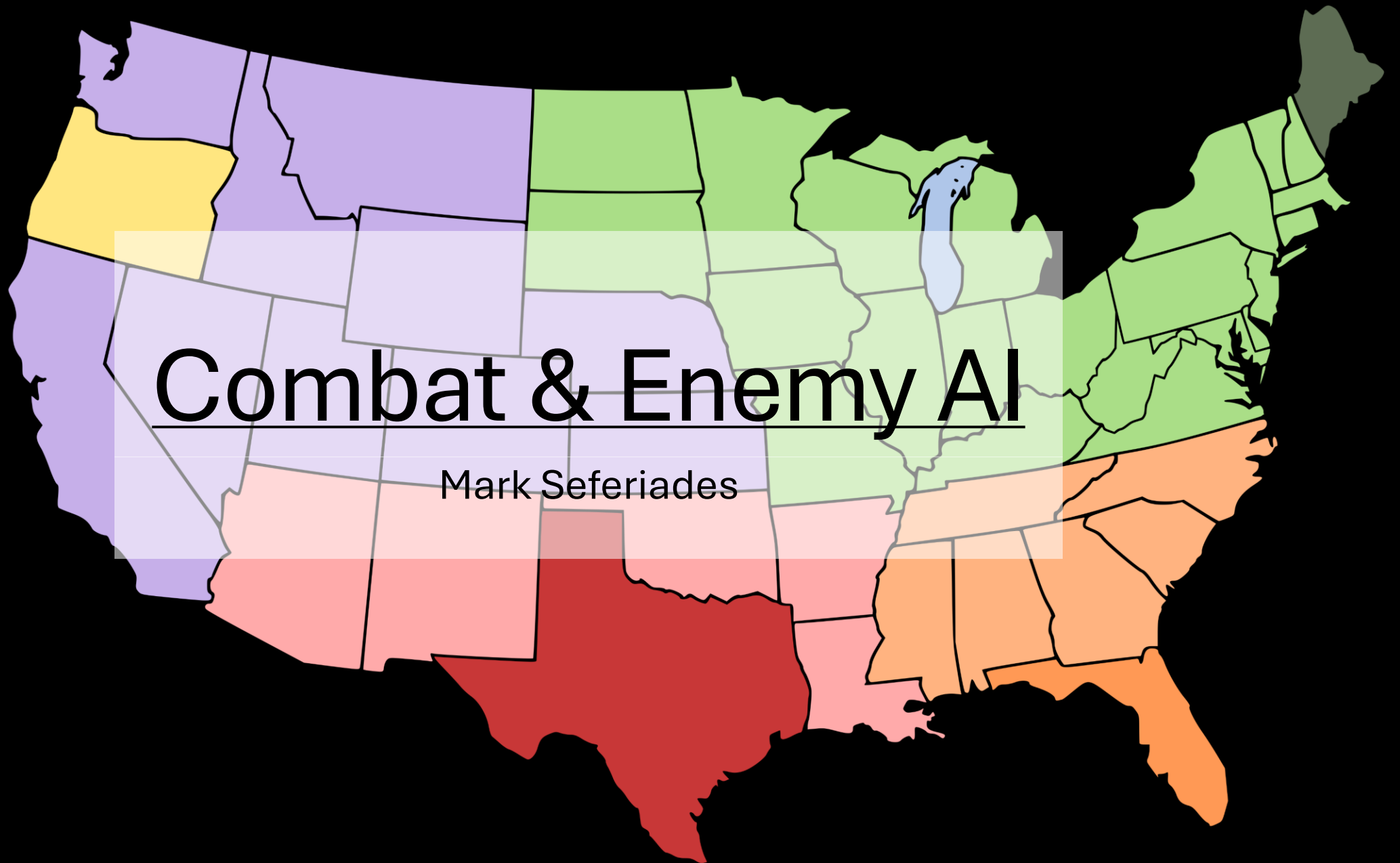
CLOUD MOD API (S3 INTEGRATION) FLOW DIAGRAM



Cloud System

CLOUD MOD API (S3 INTEGRATION) FLOW DIAGRAM

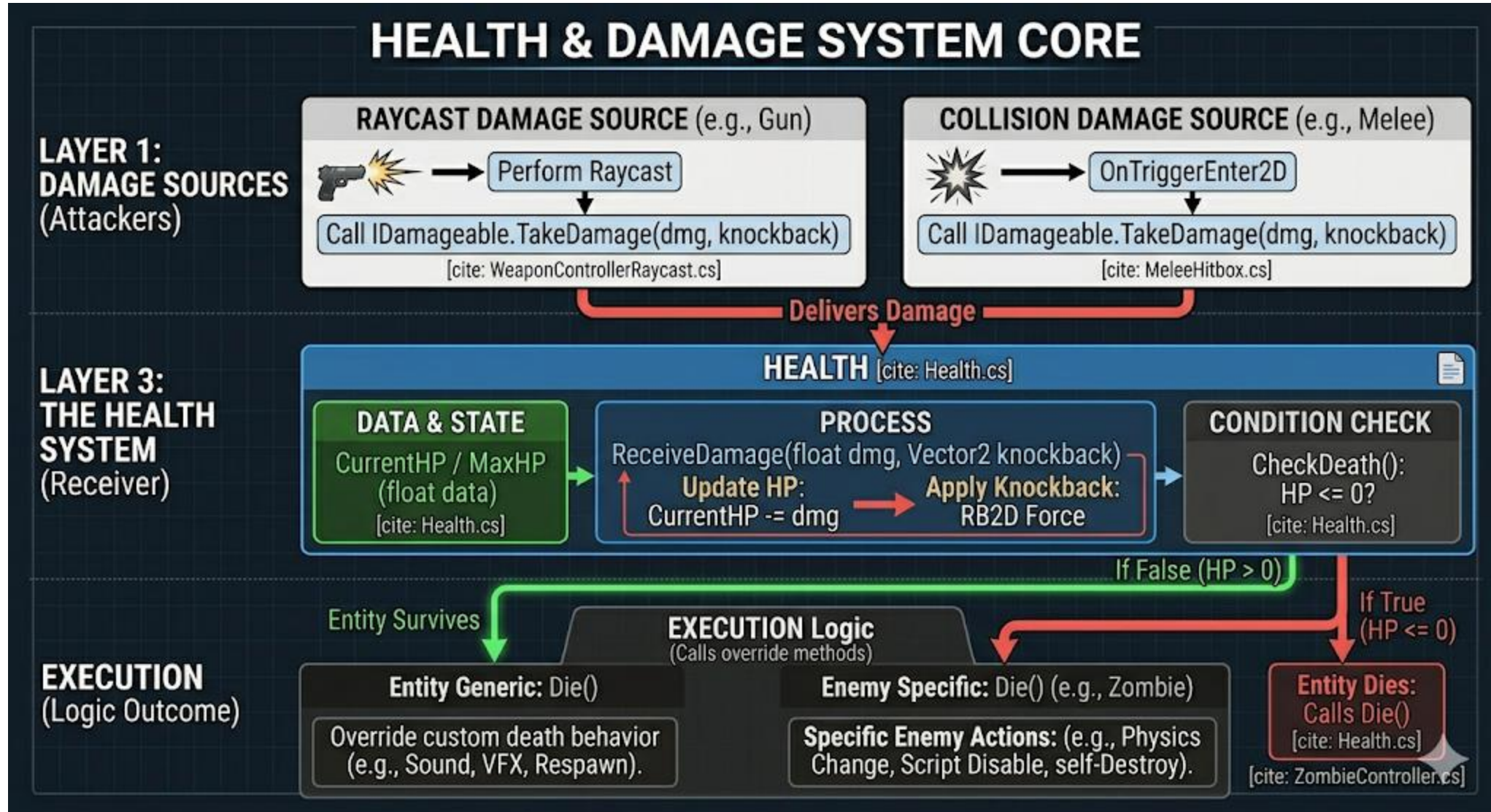




Combat & Enemy AI

Mark Seferiades

Combat System – Entities, Health and Damage



Combat System – Weapon System

Melee Weapon
Controller

Hitbox Weapons

Projectile Weapon
Controller

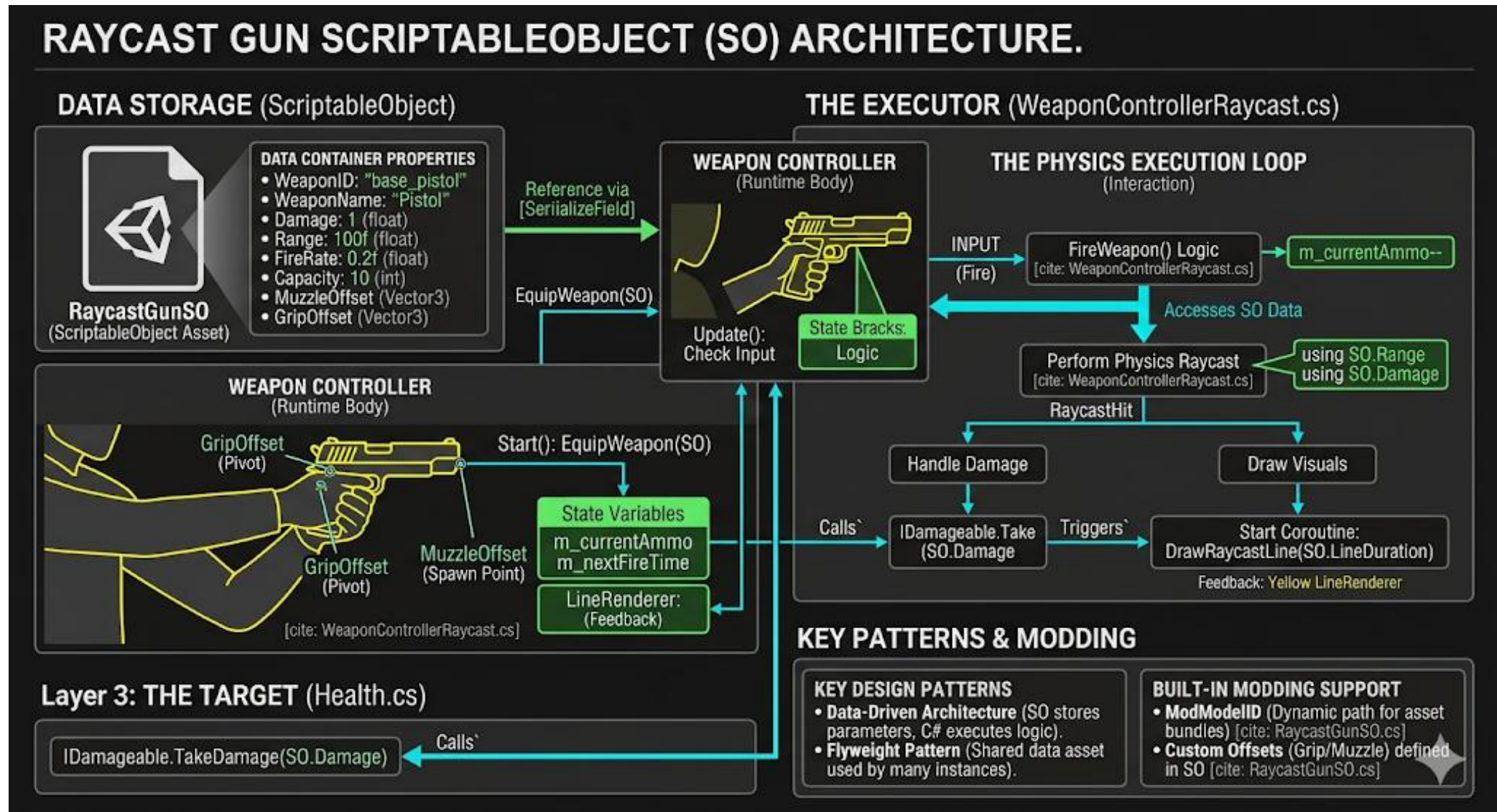
Fires game objects

Raycast Weapon
Controller

Raycast guns

Each Have their respective Scriptable Objects

Combat System – Raycast Weapon Example



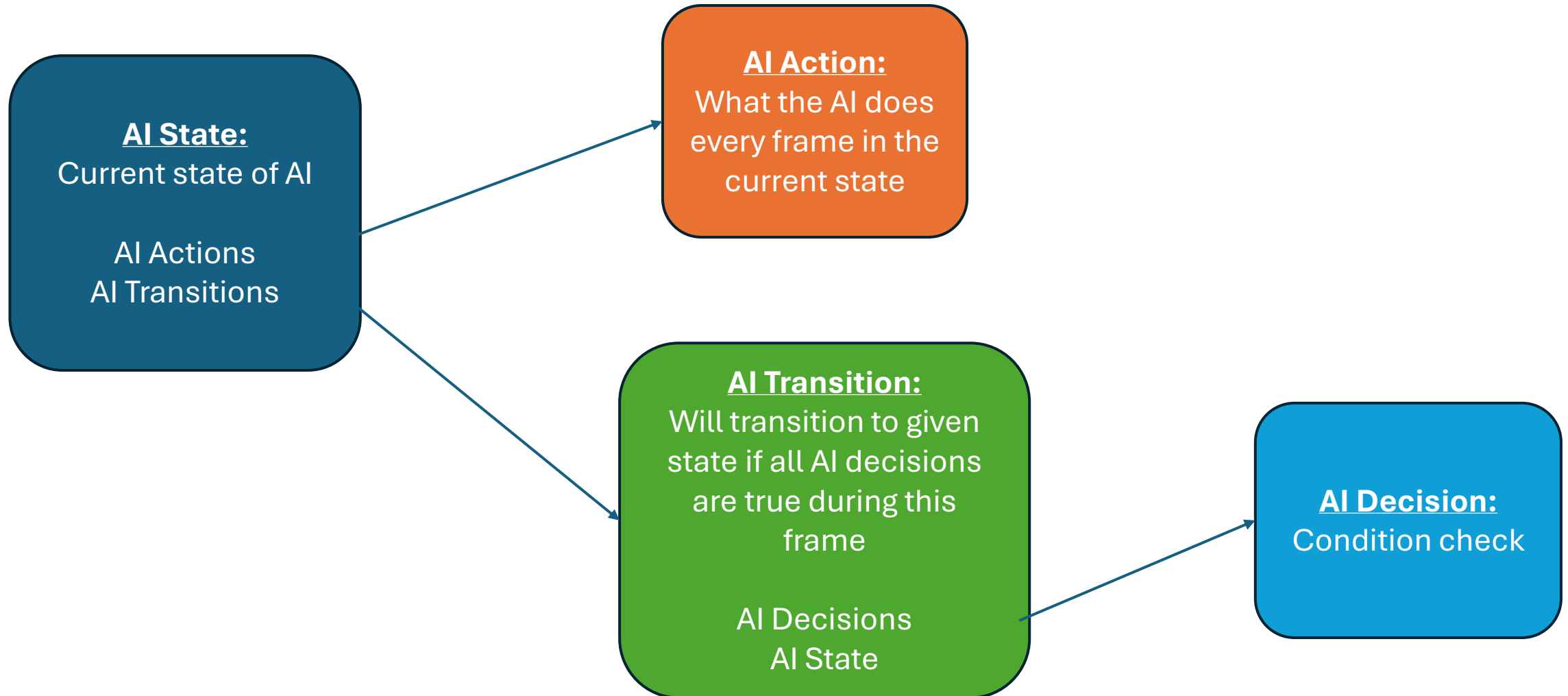
Gun Creation

Gun Name ID:	Yaa	Semi Auto	Turn On	 
Gun Damage:	10	Fired Shots	2	
Rate of Fire:	10	Weight	2	
Accuracy:	75	Attachments	String +	
DMG Range	5		Mag v -	
Reload Speed	3	Weapon Type	Pistol	
Wait Time	0	Create	Load	
Burst	0	Lock (L)		
Clip Size	30	Muzzle (M)	2.14 1.17 -5	
Ammo Type	Pistol	Grip (G)	0.97 0.98 -5	
Add Sprite	Add OBJ			
Back				

Muzzle + Grip

Gun Name ID:	Yaa	Semi Auto	Turn On	
Gun Damage:	10	Fired Shots	2	
Rate of Fire:	10	Weight	2	
Accuracy:	75	Attachments	String +	
DMG Range	5		Mag -	
Reload Speed	3	Weapon Type	Pistol	
Wait Time	0	Create	Load	
Burst	0	Lock (L)		
Clip Size	30	Muzzle (M)	2.14 1.17 -5	
Ammo Type	Pistol	Grip (G)	0.97 0.98 -5	
Add Sprite	Add OBJ			
Back				

Combat System – AI System



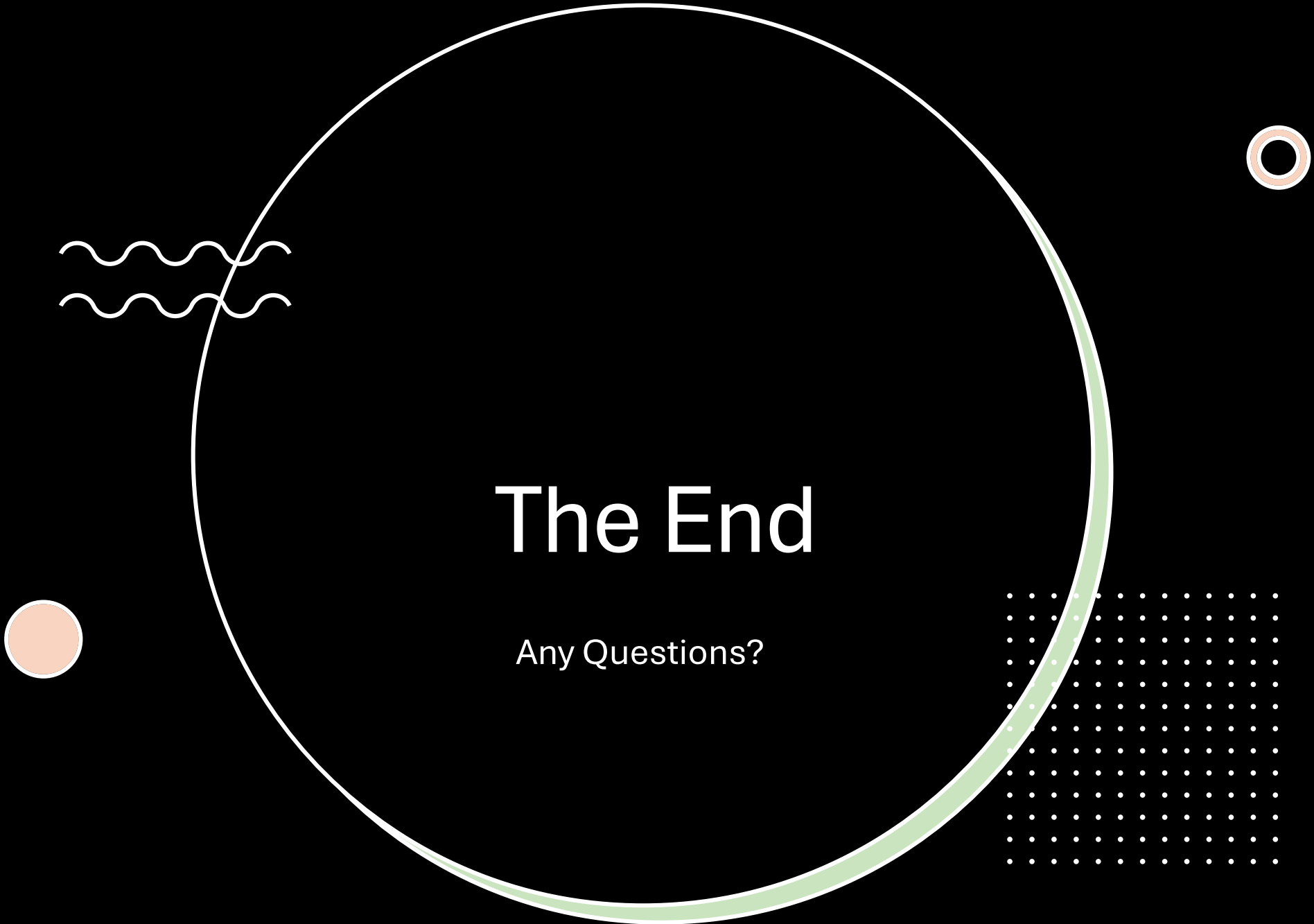
Combat System – AI System Example



Conclusion

- Survival/Action Zombie game with in-game modding, integrated with the cloud
- Worked many of the game systems to be as easily moddable as we could
- Modding can get the next generation interested in game design and programming
 - But many games lack built-in support
 - Even for some of the ones with support, they rely on external tools that can be under-documented, hard to use, and/or inconvenient
- Disclaimer: the combat system isn't fully integrated with the campaign and travel yet





The End

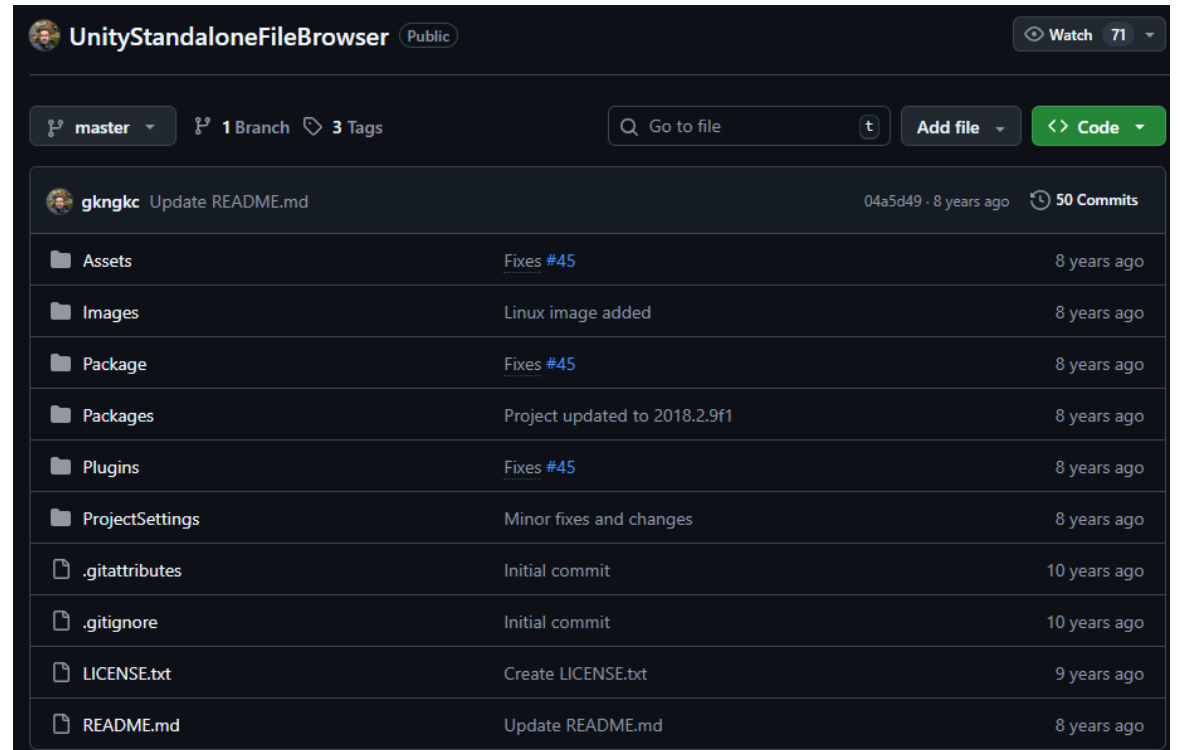
Any Questions?

OBJ Importer

By Dummiesman



By gkngkc



OBJ Importer

